

Decoding flood resilience in riverine Bangladesh: A concordance–discordance framework integrating AHP–TOPSIS and PROMETHEE II

Md. Abdullah-Al Mamun^a, Mst. Nazneen Aktar^b, Anjum Tasnuva^c, Md Tarikul Islam^d, Manoranjan Mishra^{e,f}, Abu Reza Md. Towfiqul Islam^{b,e,f,g,*}

^a Faculty of Information Technology and Communication Sciences, Tampere university, Finland

^b Department of Disaster Science and Management, Begum Rokeya University, Rangpur, 5400, Bangladesh

^c Institute of Disaster Management, Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

^d Department of Environmental Science and Disaster Management at Noakhali Science and Technology University, Sonapur 3814, Noakhali, Bangladesh

^e Department of Geography Fakir Mohan University, Vyasa Vihar, Nuapadhi, Balasore, 756089, Odisha, India

^f Department of Development Studies, Daffodil International University, Dhaka 1216 10 Bangladesh

^g Department of Earth and Environmental Science, College of Science, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul 02841, Republic of Korea

ARTICLE INFO

Keywords:

Flood resilience
Concordance -discordance analysis
AHP -TOPSIS
Riverine floodplains
Bangladesh

ABSTRACT

Flood resilience assessment in riverine systems is characterized by methodological uncertainty arising from divergent multi-criteria decision analysis (MCDA) frameworks. This study presents the first systematic concordance–discordance analysis comparing Analytical Hierarchy Process–Technique for Order Preference by Similarity to an Ideal Solution (AHP–TOPSIS; distance-based) and Analytical Hierarchy Process–Preference Ranking Organization Method for Enrichment Evaluation II (AHP–PROMETHEE II; outranking-based) approaches for flood resilience mapping across 1000 spatial locations in northern riverine Bangladesh. A total of 56 multidimensional indicators were evaluated using identical AHP–derived weights (CR = 0.153). PROMETHEE II employed Type V preference functions with indifference and preference thresholds ($q = 0.20\sigma$, $p = 0.60\sigma$). A two-level validation system was used, which comprised concordance analysis and extensive robustness testing. There was high concordance in overall resilience rankings ($p < 0.001$) and 62.0% of locations had the same resilience classification. However, large disparities in rank (mean = 112.3 positions; maximum = 464) and spatial randomness of disagreement (Global Moran I = -0.0099, $p = 0.5541$) demonstrated location-specific disagreements that could be used to prioritize intervention. Marked methodological contrasts were found in the robustness analysis. PROMETHEE II was found to have rank-reversal rates that were 80% lower and had a very high level of weight stability, whereas TOPSIS had 100% weight sensitivity, with all indicators. The indicator-level concordance of 96.4% was used to establish that 96.4% of the indicators were very highly correlated ($\Delta\rho < 0.1$) due to cumulative aggregation effects. TOPSIS offers computational simplicity suited to exploratory analysis where indicator weights are well-established and rank stability is a lesser concern, whereas PROMETHEE II's demonstrated robustness to weight uncertainty (rank-reversal and weight-sensitivity testing) makes it preferable when stakeholder-elicited weights carry inherent uncertainty and ranking stability is operationally critical, such as in high-stakes flood resilience planning.

1. Introduction

Floods are one of the most common natural disasters globally and

may have a devastating impact on human populations and nature (Mondal et al., 2021). Floods can cause a lot of destruction because of their immense magnitude, frequency, and severity; furthermore, their

* Corresponding author at: Department of Earth and Environmental Science, College of Science, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul 02841, Republic of Korea.

E-mail addresses: nazneenaktar39@gmail.com (Mst.N. Aktar), tasnuva@idm.kuet.ac.bd (A. Tasnuva), tarikul24.all@gmail.com (M.T. Islam), mmishra@sc.fmuniversity.in (M. Mishra), towfiq_dm@brur.ac.bd (A.R.Md.T. Islam).

<https://doi.org/10.1016/j.ecohyd.2026.100812>

Received 22 April 2026; Received in revised form 21 June 2026; Accepted 29 August 2026

Available online 15 September 2026

1642-3593/© 2026 European Regional Centre for Ecohydrology of the Polish Academy of Sciences. Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

economic implications and the population that they affect are ever-growing globally (Haque et al., 2024). Flooding is still one of the gravest and most frequent natural hazards in Bangladesh, which is mainly predetermined by the geomorphological location of the country in the intersection of the systems of the Ganges, Brahmaputra, and Meghna Rivers (Haque et al., 2022). Coupled with heavy monsoonal precipitation and large upstream flow from India and China, this hydrological environment predisposes the northern part, mainly the Brahmaputra Basin, to seasonal and flash floods (Dutta and Deka, 2024). The Brahmaputra Basin, which is the boundary of northeast India and northern Bangladesh, is experiencing frequent inundation episodes that lead to substantial economic losses, disruption of livelihoods, and destruction of vital infrastructure (Debnath et al., 2025). Thus, flood resilience is a critical measure that has to be evaluated to protect local communities and facilitate sustainable development.

Riverine flooding is the overflow of water in rivers or streams when the flow has surpassed the carrying capacity of the river channel and, as a result, floods adjacent floodplains and lowlands (Hossain and Mumu, 2024). Contrary to flash floods, riverine floods are experienced mostly during the monsoon season and create an unrelenting danger to human lives, livelihoods, and critical infrastructure in the north of Bangladesh (Islam et al., 2021). Extensive and severe precipitation in the upstream Brahmaputra basin, as well as transboundary inflows, is a major factor in frequent floods in the region (Islam et al., 2023; Dutta and Deka, 2024). Low topography, heavy monsoonal rainfall, and high upstream runoff lead to extensive yearly flooding of large areas in northern Bangladesh. The Brahmaputra floodplain and the greater part of the northern region are also very vulnerable to frequent riverine flooding (Rana et al., 2023). Climate change has also contributed to the increase in flood hazards through the enhancement of rainfall variability and the rise in peak discharges of river systems in the region, which have intensified the socio-hydrological effects (Ji and Wang, 2023). Historically, the northern region of Bangladesh has experienced major flood events in 1954, 1974, 1988, 1998, 2003, 2007, 2012, 2017, 2022, and 2024 (Akter et al., 2025; Hossain and Haq, 2026). These historical flood records provide a robust empirical basis for extracting multi-value reference points, which are incorporated into the prediction model to improve accuracy and validation (Debnath et al., 2025). Moreover, such records facilitate the assessment of flood impacts across different micro-level spatial units within the transboundary riverine landscape. Kurigram District, in particular, experiences flooding almost every year, resulting in severe socio-economic disruption and extensive infrastructural damage (Islam et al., 2023). Therefore, a comprehensive framework for assessing flood risk that integrates both physical and socio-economic dimensions is critically important for effective flood management and resilience planning in the northern region of Bangladesh.

Given this transboundary flood risk, there is increasing demand for spatially obvious flood resilience assessments that can support evidence-based disaster risk reduction (DRR), infrastructure planning, and resource allocation. Flood resilience, defined as the capacity of communities to absorb, adapt to, and recover from flood impacts, extends beyond physical protection measures to include socio-economic conditions, institutional capacity, and preparedness systems (Ansari et al., 2022). For decision-makers, understanding how resilience varies across space is essential for prioritizing interventions in highly vulnerable riverine environments (Chen et al., 2024).

MCDA has become a widely adopted decision-support tool for flood risk and resilience assessment because it enables the integration of heterogeneous indicators across environmental, social, economic, and infrastructural domains (Aroca-Jiménez et al., 2023). Nonetheless, the methods of MCDA vary significantly in the way they combine information and convert the performance of indicators into ratings, and these differences may cause conflicting management decisions. These methodological differences lead to the critical question of whether, for environmental planners and policy practitioners, the various MCDA methodologies can deliver consistent and robust flood resilience

assessments (Rana et al., 2023).

MCDA methods based on distance, like the Technique of Order of Preference by Similarity to Ideal Solution (TOPSIS), are methods for ranking alternatives based on their relative closeness to an ideal and an anti-ideal reference point. These are computationally efficient and straightforward-to-understand methods that can make trade-offs between criteria using compensatory aggregation (Rizalighadi et al., 2026). Therefore, TOPSIS has been extensively used to manage the environment, prioritize infrastructure, and map flood vulnerability (Dutta and Deka, 2024). Nevertheless, the distance-based techniques have been found to be affected by the uncertainty of weight and the reverting of ranks in cases where the composition of the alternatives alters (Wang and Triantaphyllou, 2008).

Instead, outranking-based methods, especially PROMETHEE II, are based on a pairwise dominance of the alternatives, instead of a distance to an ideal solution (Plataridis and Mallios, 2024). In PROMETHEE II, preference and indifference thresholds are included to show tolerance for variability in performance levels by the decision-maker, thus preventing excessive compensation in relation to the criteria. This structure has more resistance to reversal of ranks and a higher level of robustness in the face of uncertainty, which is of particular importance to a high-stakes environmental decision-making process (Coquelet et al., 2025).

Despite the vast and extensive use of both TOPSIS and PROMETHEE II in environmental assessment, there have been limited systematic empirical comparisons of the performance, robustness and spatial behavior of both in flood resilience assessment, especially in highly vulnerable riverine settings in the riverine region of Bangladesh. Current literature tends to compare processes with correlation coefficients only but does not consider spatial consistency, rank consistency, or sensitivity to parameter uncertainty, which are all important to management reliability (Taoukidou et al., 2025). Although the number of studies on flood resilience is increasing, the application of concordance and discordance analysis based on AHPTP-using AHP-TOPSIS and AHP-PROMETHEE II in the northern part of Bangladesh is limited, which implies a significant and obvious research gap.

To fill this research gap, the current study will be a comprehensive, empirical comparison of AHP-TOPSIS with AHP-PROMETHEE II to evaluate spatial flood resiliency in Kurigram District, northern Bangladesh. Both methods are compared using a harmonized dataset of 1000 spatial locations and 56 indicators given weighted by AHP under the same conditions. It suggests a two-level validation model, consisting of (i) concordance validation of rankings and spatial congruence and (ii) robustness validation of a broad scope, including rank reversal, sensitivity to weight, normalization effects, spatial structure and parameterization of preferences.

Flood risk refers to the potential adverse consequences arising from the interaction of flood hazards, exposure, and vulnerability (Islam et al., 2025). Flood vulnerability reflects the susceptibility of exposed systems to flood impacts due to their sensitivity and limited adaptive capacity. Flood resilience represents the capacity of communities and systems to anticipate, absorb, adapt to, and recover from flood impacts across social, economic, infrastructural, institutional, and ecological dimensions (Ansari et al., 2022; UNDRR, 2015; Chen et al., 2024). The Composite Flood Resilience Index (CFRI) developed in this study is a dimensionless, spatially ranked measure derived from AHP-weighted TOPSIS and PROMETHEE II using 56 indicators. It measures relative levels of resilience, but it doesn't directly measure flood risk or vulnerability, which are two different but related ideas.

The study is guided by four research questions addressing the following:

- (i) the degree of agreement between distance-based and outranking-based flood resilience rankings;
- (ii) the spatial distribution of methodological concordance and discordance;

- (iii) comparative robustness under uncertainty and data perturbation;
- (iv) practical implications for MCDA method selection in flood risk management.

This research paper discusses four research questions that are connected to each other and contribute to the knowledge of methodological concordance, spatial patterns of agreement, robustness properties, and practical implications for selecting MCDA methods in flood resilience assessments. So far, this is the first systematic comparison of these two popular MCDA methods to specifically assess flood resilience at this spatial resolution (1000 locations) with such a rich set of indicators (56 indicators in 6 dimensions) and provides empirical results on how well they agree, how they disagree, and how robust they are. This study introduces a novel contribution to the current body of literature by providing practical advice on which methodology to use in MCDA to choose the optimal method to use when planning flood resilience, prioritizing resources, and adapting to climate variability in the vulnerable riverine floodplains. Instead of promoting a universally better approach, it indicates that TOPSIS can be used in an exploratory setting or in a resource-constrained setting, whereas PROMETHEE II should be used in high-stakes decisions that need to be made in an

uncertain environment. The conclusions encourage context-based MCDA to be used in flood resilience planning in a systematic way.

2. Methodology

2.1. Study area description

The Kurigram District (25°23' - 26°14' N, 89°27' - 89°54' E) is located in one of the most flood-susceptible areas in Bangladesh, in the active floodplains of the Brahmaputra (Jamuna), Teesta and Dharla Rivers (Fig. 1). The district experiences severe and frequent flooding due to transboundary river flows, heavy monsoons, and dynamic geomorphology, including channel migration, char formation, and river-bank erosion. The population supported by five sub-districts, Nageshwari, Kurigram Sadar, Ulipur, Chilmari, and Raumari, with over two million inhabitants, is mainly reliant on unpredictable agricultural lifestyles. Annual rainfall exceeds 2400 mm, with 80% concentrated in the monsoon season (June-September). The consequences of these conditions are almost annual flooding, typified by sustained and rapid floods; embankment overflows; and congested drainage.

Flood hazard zonation (Fig. 1) integrates three datasets: (i) (i) 500-

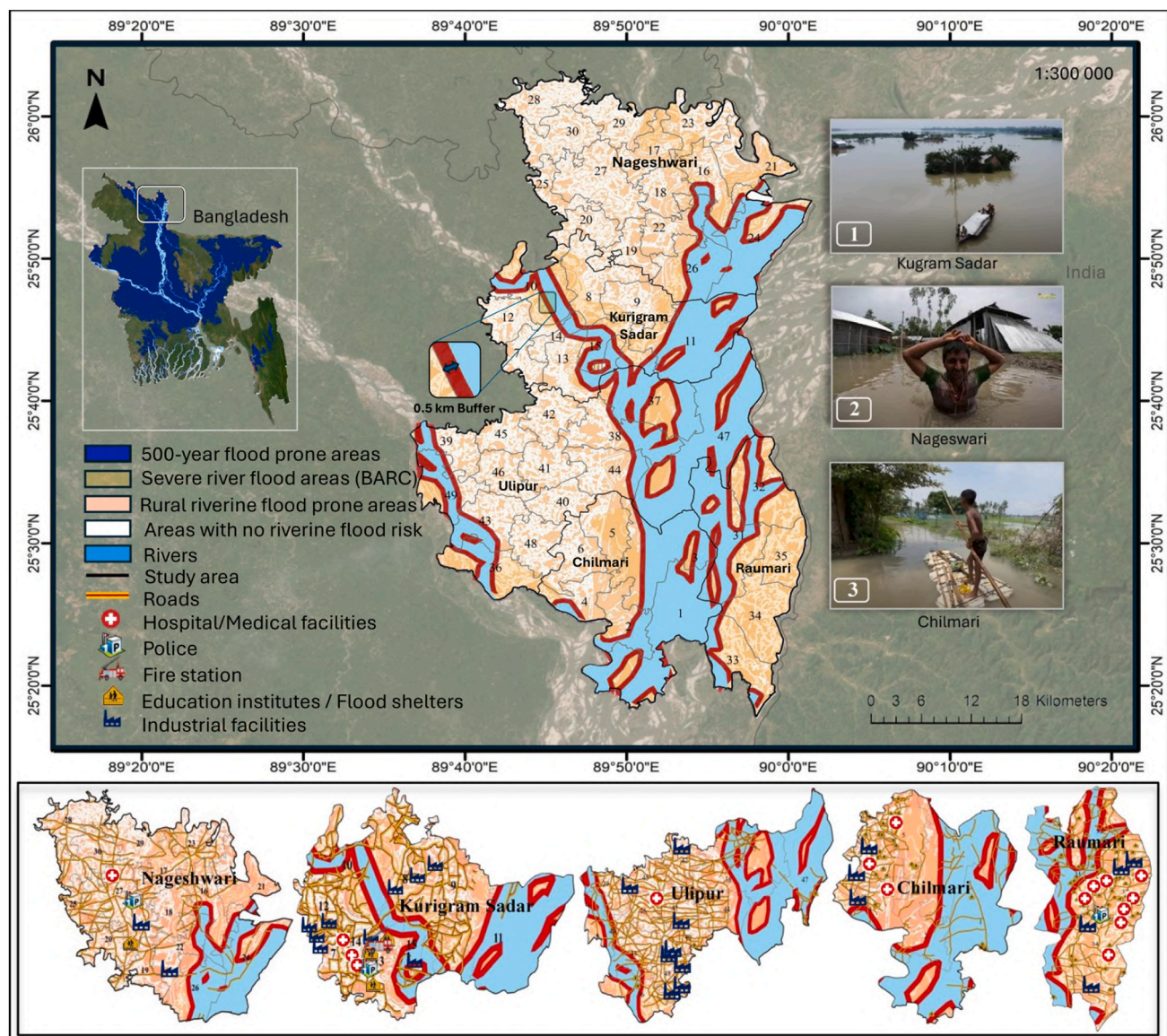


Fig. 1. Study area map and flood hazard zonation for Kurigram District, Bangladesh.

year flood-prone zones from the European Commission Joint Research Centre global modelling within 0.5 km of major rivers; (ii) areas at risk of severe river flooding, which is from the Bangladesh Agricultural Research Council; and (iii) areas at risk of riverine flooding, determined by Copernicus GLO-30 DEM-based terrain analysis (slopes $\leq 2\%$). The risk of flooding is concentrated at the Brahmaputra-Teesta confluence. Field observation shows that there are extensive effects, such as submerged settlements and disrupted transport infrastructure. In cities, critical facilities are concentrated, leading to large access disparities during floods.

Fig. 1 documents these conditions across three complementary views: the main panel overlays critical infrastructure (hospitals, fire stations, police, and education/shelter facilities) onto the flood hazard zonation described above; inset photographs (1–3) record observed flooding at Kurigram Sadar, Nageshwari, and Chilmari, illustrating the submerged settlements and disrupted transport noted above; and the lower panel provides upazila-level detail for each of the five sub-districts (Nageshwari, Kurigram Sadar, Ulipur, Chilmari, and Raumari), showing the spatial relationship between road networks, flood-prone zones, and facility locations underlying the access disparities described above.

2.2. Data collection and sampling strategy

Primary data were collected using structured household survey methods as a sampling strategy. Structured household surveys were used to collect primary data in 49 unions (Supplementary Table S3) from June to October 2024 and were organised by the Rangpur Dinajpur Rural Service (RDRS) in Bangladesh. The formula was used to calculate the sample size using Slovin ($n = N / (1 + N \times e^2)$) (Cochran, 1977), where a population of 29,760 households (N) and a 3% level of error ($e = 0.03$) yielded a target sample of ~ 1072 households (rounded to 1000). Trained enumerators conducted surveys using KoboToolbox for real-time standardised data collection with validation and quality control. A harmonised 5-point ordinal scale (Very Low to Very High) was used to measure all indicators to ensure consistent directionality across the 56 indicators that constitute the six dimensions of resilience. Household responses were individually geocoded to their household location using ArcGIS 10.8 (WGS 1984, UTM Zone 46 N), yielding 1000 unique spatial points; union boundaries (Supplementary Table S3) were used only to stratify sampling across the district and to report descriptive context, with all subsequent MCDA and spatial analyses (TOPSIS, PROMETHEE II, kriging, LISA) conducted at the individual household-point level. Flood hazard layers, infrastructure points, and topography were included in spatial datasets derived from the Copernicus Programme GLO-30 Digital Elevation Model (DEM).

2.3. Multi-Dimensional flood resilience framework

Flood resilience was conceptualised in six dimensions, each inter-dependent, with 56 indicators (Table 1). The framework consists of six dimensions, each with its corresponding weight (w) and leading indicators (i): socio-cultural ($i = 11$; $w = 0.3285$; led by EMC 0.040, SOG 0.037), economic ($i = 6$; $w = 0.1435$; led by INS 0.035, MSM 0.030), infrastructural-physical ($i = 13$; $w = 0.2274$; led by WEES 0.022, INF 0.021), organisational-institutional ($i = 8$; $w = 0.1158$; led by INA 0.019, OPA 0.018), hydraulic ($i = 8$; $w = 0.0890$; led by STA 0.013, UNA 0.013), and ecological ($i = 10$; $w = 0.0959$; led by REK 0.013, BNA 0.012). The descriptive indicators are shown in more detail in Supplementary Table S1.

These dimensions interact dynamically: social and economic capacities influence infrastructure effectiveness, hydraulic and ecological conditions determine hazard magnitude, and institutional governance coordinates system-wide responses. The framework consists of 45 benefit and 11 cost indicators, with weights ranging from 0.011 to 0.040 ($CV = 0.387$), and flood resilience is conceptualised as an emergent, adaptive system outcome (Supplementary Table S1).

Table 1

Multi-Dimensional Framework for Flood Resilience Assessment in Kurigram District, Bangladesh.

Dimension	Indicators (n)	Benefit (+)	Cost (-)	Total Weight	Indicator by Weight (Top 3)
Socio-Cultural	11	10	1	0.3285	EMC (0.040), SOG (0.037), PVC (0.035)
Economic	6	4	2	0.1435	INS (0.035), MSM (0.030), FVP (0.023)
Infrastructural-Physical	13	11	2	0.2274	WEES (0.022), INF (0.021), TRP (0.019)
Organizational-Institutional	8	8	0	0.1158	INA (0.019), OPA (0.018), SAO (0.016)
Hydraulic	8	5	3	0.0890	STA (0.013), UNA (0.013), WDNr (0.012)
Ecological	10	7	3	0.0959	REK (0.013), BNA (0.012), ACHMP (0.011)
Total	56	45	11	1.0000	

Directionality was assigned based on the established literature relationship between each indicator and flood resilience outcomes. Indicators positively associated with coping, adaptive, or recovery capacity were classified as benefit (+), while indicators reflecting heightened exposure, vulnerability, or hazard susceptibility were classified as cost (-). The specific theoretical basis for each of the 56 indicators is cited individually in Supplementary Table S1.

2.4. Proposed methodological framework for flood resilience assessment

The proposed methodological framework combines expert-based MCDA and robustness validation of 1000 spatial sites using four elements: weight derivation, parallel ranking, comparative evaluation, and advanced validation (Fig. 2). To compute indicator weights, AHP was used. Pairwise comparisons of 56 indicators using Saaty's 1–9 scale were given by five flood experts and aggregated using the geometric mean. To check consistency, the Consistency Ratio ($CR \leq 0.167$) was used, and the normalized weight vector ($\sum w_j = 1.000$) was applied to both MCDA approaches in the same manner. Second, AHP-TOPSIS and AHP-PROMETHEE II generated parallel rankings (1000 locations $\times$ 56 indicators) using distance-based aggregation (closeness coefficients) and pairwise outranking with Type V preference functions (Φ_{net}), respectively.

Third, methodological concordance and strength were evaluated at a second level. Tier 1 measured agreement in terms of rank correlation measures (Spearman's ρ , Kendall's τ , Pearson's r) and mapping of spatial agreement. Tier 2 measured robustness using rank reversal, weight sensitivity, spatial autocorrelation, normalization sensitivity, and preference function testing. Lastly, further validation measures, such as ROC-based concordance prediction, indicator-level analysis, and spatial hotspots, identified the method(s) that are most stable and reliable to assess operational flood resilience.

2.5. AHP-Based indicator weighting

Before employing the Analytic Hierarchy Process (AHP), the authors explored the entire body of existing subjective weighting techniques, including BWM (Rezaei, 2015), SWARA (Keršulienė et al., 2010), DEMATEL (Gabus and Fontela, 1972), SMART/SWING (Edwards and Barron, 1994), SIMOS (Figueira and Roy, 2002), and direct point allocation methods. Both have unique pros and cons related to the simplicity

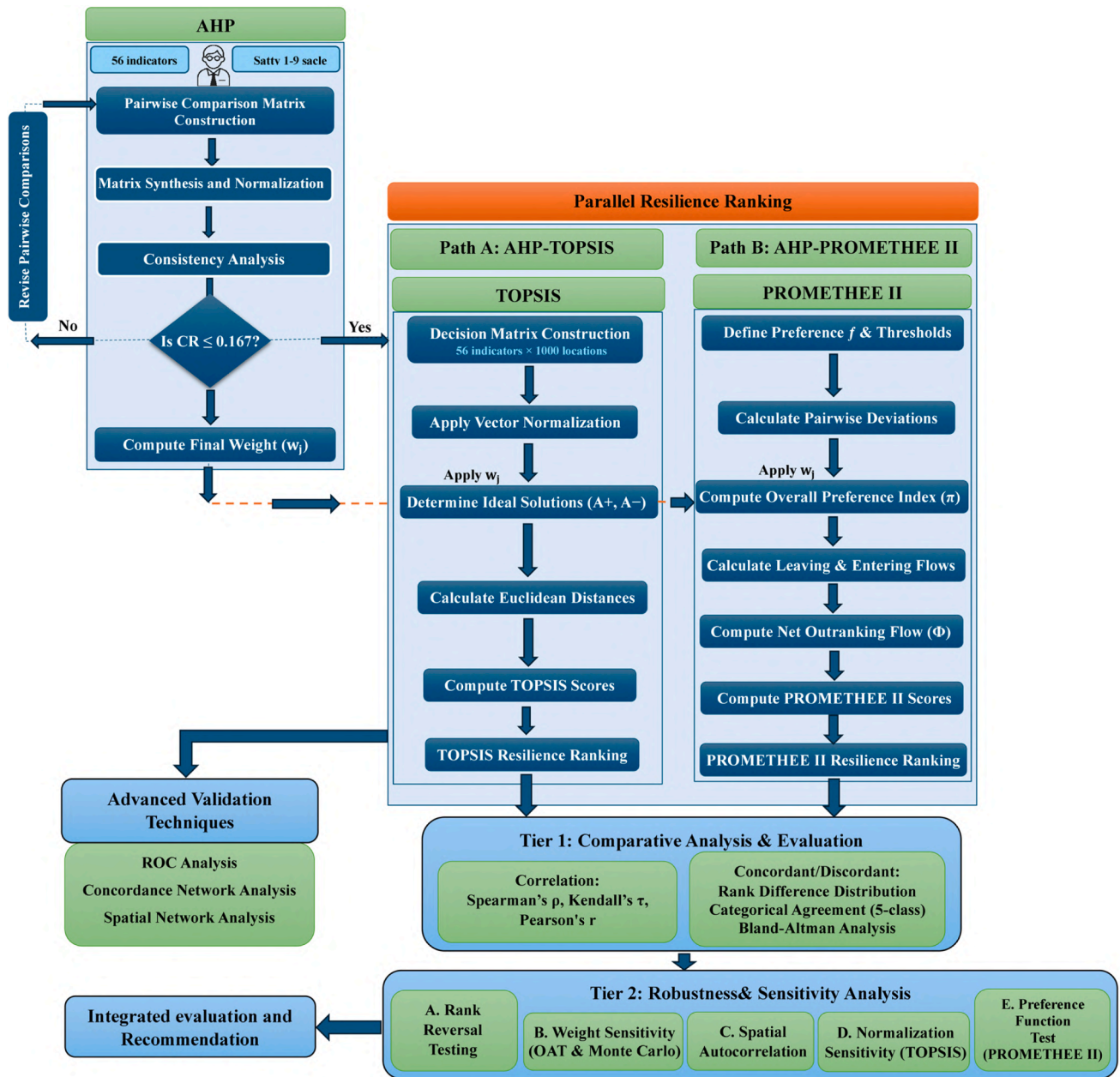


Fig. 2. Methodological framework for comparative flood resilience assessment using AHP-TOPSIS and AHP-PROMETHEE II across 1000 locations.

of elicitation, consistency control, and suitability for a large set of indicators (Kizielewicz et al., 2024). However, BWM and SWARA do not include the ability to decompose the indicators into a set of hierarchical indicators, as needed to manage 56 cross-dimensional indicators across six resilience domains. DEMATEL is convenient for causal mapping, while SIMOS is convenient for ordinal prioritization but can be used to generate eigenvector-based weights for structured group aggregation. Considering the hierarchical, multi-expert and multi-dimensional nature of this study, the Fuzzy AHP method was chosen due to its abilities in hierarchical structuring, consistent ratio and geometric-mean group aggregation, which has been proved to be valid in the flood and disaster risk assessment literature (Dutta and Deka, 2024; Ekmekcioğlu et al., 2021; Chen et al., 2024).

The AHP was used to obtain indicator importance weights through structured expert elicitation. Pairwise comparisons of all 56 indicators were made by five specialists in flood risk and disaster management using Saaty's 1–9 scale (Saaty, 1987), where 1 denotes equal importance and 9 denotes extreme importance. The geometric mean was used to

aggregate individual expert ratings to form a consolidated 56×56 pairwise comparison matrix (Supplementary Table S5). Classical (crisp) AHP was applied throughout, without fuzzification of pairwise judgments or a defuzzification step, to preserve direct interpretability of the Saaty 1–9 scale and the consistency-ratio diagnostics described below.

The CR was used to measure matrix consistency, calculated as $CR = CI/RI$, where CI is the Consistency Index based on the principal eigenvalue (λ_{max}) and RI is the Random Index based on the matrix size ($n = 56$). MCS (10,000 randomly generated Saaty matrices) yielded an empirical RI of $1.617 (\pm 0.0003 \text{ SE})$. This is very close to the interpolated value provided by Donegan and Dodd (1991) ($RI = 1.615$; 0.14% difference), indicating that the estimation is reliable (see Supplementary Materials).

In line with recent applied research (Ekmekcioğlu et al., 2021), $RI = 1.8$ was used, yielding a CR of 0.153. This is considered acceptable for large matrices ($n > 15$), since it is difficult to obtain $CR \leq 0.10$ due to cumulative comparison uncertainty (Saaty, 1987). Other RI values produced CR values of 0.168 to 0.170 (Supplementary Table S5). Final

normalized weights (principal eigenvector) ranged from 0.0069 to 0.0397 (mean = 0.0179; Supplementary Fig. S3). In both TOPSIS and PROMETHEE II analyses, these weights were used identically to ensure consistency in indicator importance across the two approaches. AHP was selected among subjective weighting approaches because it provides a structured and transparent framework suitable for this study's 56-indicator flood resilience assessment. First, its pairwise comparison mechanism (Saaty, 1987) decomposes a complex weighting problem into manageable expert judgments, making it more practical than methods requiring direct scoring of all indicators simultaneously (e.g., SMART). Second, AHP incorporates a quantitative consistency assessment through the consistency ratio (CR = CI/RI), enabling evaluation of expert judgment reliability rather than relying solely on subjective agreement. Third, AHP remains one of the most widely applied weighting techniques in flood-risk and disaster resilience MCDA studies (e.g., Debnath et al., 2025; Dutta and Deka, 2024), allowing comparison with established regional evidence.

Despite these advantages, AHP has identified limitations, particularly when dealing with large numbers of criteria. For 56 indicators, a full comparison matrix requires 1540 pairwise judgments per expert, which may exceed normal cognitive processing capacity and reduce consistency (Liu et al., 2020). To address this challenge, the elicitation process was conducted through hierarchical decomposition, two-stage expert sessions, and interim CR feedback. Although a stricter CR threshold is often recommended, achieving very low inconsistency levels becomes increasingly difficult for large matrices; therefore, a threshold of $CR \leq 0.153$ was adopted as a methodological compromise consistent with previous flood resilience studies (Ekmekcioğlu et al., 2021; Dutta and Deka, 2024).

AHP-derived weights are also influenced by expert selection and elicitation design. To minimize this uncertainty, the robustness of the weighting framework was evaluated using one-at-a-time (OAT) sensitivity analysis, Monte Carlo simulation, and the Fuzzy AHP approach (Sections 2.7.2.2–2.7.2.3). These analyses quantify the influence of weighting uncertainty on the final rankings rather than treating expert-derived weights as fixed assumptions. While alternative approaches such as the Best-Worst Method (BWM), which reduces comparison requirements (Rezaei, 2015), and DEMATEL, which considers inter-criteria relationships (Gabus and Fontela, 1972), provide potential advantages, their application was beyond the current study scope and is suggested for future comparative investigations.

To further evaluate the robustness of the weighting strategy, objective weighting methods based on entropy and CRITIC (Diakoulaki et al., 1995) were calculated independently from expert preferences. These objective weights were integrated with AHP weights through multiplicative normalization, generating four alternative schemes (entropy-only, CRITIC-only, AHP $\times$ entropy, and AHP $\times$ CRITIC). The resulting rankings were subsequently compared with the original AHP-based rankings for both AHP-TOPSIS and AHP-PROMETHEE II models (Supplementary Table S8).

2.6. Dual MCDA implementation for resilience ranking

2.6.1. AHP-TOPSIS

TOPSIS ranks locations by comparing their indicator profiles to the best and worst possible solutions, based on the most extreme data points (Hwang and Yoon, 1981), using a distance-based method (Brans and Vincke, 1985).

The algorithm has four steps: (i) normalize the 1000×56 decision matrix as a vector; (ii) multiply the data by weights from AHP (see Supplementary Table S4); (iii) identify the ideal (A^*) and anti-ideal (A^-) solutions based on their benefit or cost; and (iv) calculate Euclidean distances (S^+ , S^-) to get the TOPSIS CFRI (closeness coefficient C^*) using the given equation. Detailed steps are in Supplementary Materials (ST3).

2.6.2. Resilience ranking with AHP-TOPSIS

In this paper, the AHP-TOPSIS model identified the Composite Flood Resilience Index (CFRI) as the TOPSIS CFRI (closeness coefficient C_i^*). CFRI is a continuous unitless resilience measure where higher values indicate greater resilience, consistent with established AHP-TOPSIS practices (Moghadass et al., 2019). The 1000 sites were listed in order of decreasing CFRI, with higher CFRI values ranked first. To facilitate interpretation and decision making, the continuous CFRI distribution was divided into five ordinal resilience classes, namely, Very Low, Low, Moderate, High, and Very High, with the use of the natural breaks (Jenks) classification technique. The technique identifies the thresholds between the classes in order to maximize within-class homogeneity and between-class heterogeneity to retain the original structure of the resilience score distribution.

2.6.3. AHP-PROMETHEE II

PROMETHEE II (Preference Ranking Organization Method for Enrichment Evaluation) ranks alternatives through a systematic pairwise outranking analysis based on preference functions that model decision-maker tolerance for performance differences (Brans and Vincke, 1985). Unlike TOPSIS's distance-based compensatory aggregation, PROMETHEE II evaluates each location pair individually across all indicators, implementing semi- to non-compensatory logic through preference function threshold structures (Brans and Vincke, 1985; Schär et al., 2025).

Type V preference functions employ indifference and preference thresholds calibrated as $q = 0.20\sigma$ and $p = 0.60\sigma$, where σ = indicator standard deviation (Supplementary Table S7). For each location pair (a , b) and indicator j , the performance difference is computed as $d_j = g_j(a) - g_j(b)$ for benefit indicators and $d_j = g_j(b) - g_j(a)$ for cost indicators (Eq. (1)) and then mapped to a preference degree $P_j(a, b) \in [0, 1]$ using the Type V function (Eq. (2)). Pairwise preferences are aggregated using AHP-derived weights across all $m = 56$ indicators to form the multicriteria preference index $\pi(a, b)$ (Eq. (3)), from which the positive and negative outranking flows over $n = 1,000$ locations quantify dominance $\Phi^+(a)$ (Eq. (4)) and weakness $\Phi^-(a)$ (Eq. 5); the final PROMETHEE II score is given by the net flow $\Phi_{\text{net}}(a) = \Phi^+(a) - \Phi^-(a)$ (Eq. (6)).

$$d_j = \begin{cases} g_j(a) - g_j(b), & \text{benefit indicator} \\ g_j(b) - g_j(a), & \text{cost indicator} \end{cases} \quad (1)$$

$$P_j(a, b) = \begin{cases} 0, & d_j \leq q_j \\ \frac{d_j - q_j}{p_j - q_j}, & q_j < d_j < p_j \\ 1, & d_j \geq p_j \end{cases} \quad (2)$$

$$\pi(a, b) = \sum_j w_j P_j(a, b) \quad (3)$$

$$\Phi^+(a) = \frac{1}{n-1} \sum_{x \neq a} \pi(a, x) \quad (4)$$

$$\Phi^-(a) = \frac{1}{n-1} \sum_{x \neq a} \pi(x, a) \quad (5)$$

$$\Phi_{\text{net}}(a) = \Phi^+(a) - \Phi^-(a) \quad (6)$$

2.6.4. Resilience ranking with AHP-PROMETHEE II

The ranking of the locations in PROMETHEE II is obtained by sorting each location in descending order of the Φ_{net} values. Such net flows form the CFRI of AHP-PROMETHEE II and are used as the basis of resilience classification. Just like TOPSIS, the continuous Φ_{net} distribution was categorised into five resilience classes using natural breaks (Jenks) classification. The full pairwise outranking model evaluates approxi-

mately $n(n-1)$ ordered pairs (~ 1 million for $n = 1000$) across 56 indicators, requiring ~ 56 million preference function evaluations and computational optimization. Because of this high computational cost, especially for large-scale sensitivity computations, calculations were run in Google Colab on a GPU-enabled Google Compute Engine instance (NVIDIA Tesla T4; CUDA 12.4).

2.7. Proposed two-tier evaluation: concordance analysis and robustness testing

2.7.1. TIER 1: basic concordance analysis

To measure methodological concordance, a comparative analysis was conducted that involved rank correlation, positional displacement, and categorical agreement. The two procedures were used on the same input data such that differences observed are inherent in the procedures. Rank correlation used three measures, including Spearman (r) (monotonic association), Kendall (τ) (pair concordance) and Pearson (r) (linear associations between scores) (Sheskin, 2011). All correlations were checked at a significance level of $\alpha = 0.001$. Rank difference analysis was used to measure the positional displacement $\Delta \text{rank} = |\text{Rank (TOPSIS)} - \text{Rank (PROMETHEE II)}|$ of each location; descriptive statistics were mean, median, standard deviation, maximum, and percentile distributions. Categorical concordance grouped locations into four groups of Strong Concordance (50), Moderate Concordance (51–100), Mild Discordance (101–200), and Strong Discordance (>200) based on the practical decision-making conditions where top 5, 10 and 20 percent cutoffs are applied to allocate resources. Concordance between categories of resilience Agreement between categories of resilience was assessed as the precise agreement across the categories of resilience (Very Low, Low, Moderate, High, Very High) using Jenks natural breaks (Jenks, 1967).

2.7.2. TIER 2: comprehensive robustness testing

2.7.2.1. Test A: rank reversal testing. Rank reversal analysis (Zanakis et al., 1998) assesses the stability of MCDA through systematic elimination of options, which is a typical comparative method of evaluating whether the rank order will stay the same. Three removal scenarios were implemented: Top 10% (top 100 ranks), Bottom 10% (ranks 901–1000), and Middle 10% (ranks 451–550). In every scenario, the 100 locations mentioned were left out and both MCDA procedures were re-calculated on the 900 locations with the same parameters. The resultant rankings were compared to the original rankings. The rank reversal percentage (RR%) was calculated as the percentage of pairwise rank order reversals of the 404,550 possible pairs of alternatives. The smaller the RR% values, the more stable the ranking will be. Besides, the rank correlation coefficient (ρ) of Spearman between the original and recalculated rankings was used as an auxiliary, quantitative stability measure.

2.7.2.2. Test B1: weight sensitivity analysis (OAT method). One-At-a-Time (OAT) sensitivity analysis (Triantaphyllou and Sánchez, 1997) is an assessment of ranking the response to individual weight perturbations in the presence of other variables, and as such, is a baseline robustness test (Wienckowski and Salabun, 2025). The weight on each of the 56 indicators was then varied in six levels ($\pm 10\%$, $\pm 20\%$, $\pm 30\%$) subject to the sum-to-one constraint. Each of the 336 perturbation cases was recalculated with the two MCDA methods and then compared to the baseline rankings with the Spearman rank correlation coefficient (ρ). Indicators have been categorized as Robust ($\rho \geq 0.95$ and above), Moderately Sensitive ($0.90 \leq \rho < 0.95$), or Highly Sensitive.

2.7.2.3. Test B2: weight uncertainty analysis (MCS). MCS comprises the spreading of the uncertainty of the weights with 10,000 trials (Saltelli et al., 2004). Each trial sampled 56 wt of normal distributions with centers that were the AHP weights and standard deviation that was a

measure of uncertainty of the measurement (5–10% of the weight value). Normalized perturbed vectors were created with a sum of 1.0 and both methods computed rankings. Measures of uncertainty were mean rank, median, 95% confidence interval, and rank volatility. Narrow confidence intervals indicate robust rankings insensitive to weight uncertainty.

2.7.2.4. Test C: spatial autocorrelation analysis. Global Moran's I and Local Indicators of Spatial Association (LISA) were used to evaluate spatial autocorrelation. These methods help determine whether geographic clustering or spatial randomness exists in rank differences between AHP-TOPSIS and AHP-PROMETHEE II (Anselin, 1995; Mamun et al., 2024). Section 2.7 quantifies this analysis as spatial dependence in rank differences within the study area. The rank differences are then tested for significance ($\alpha = 0.05$) to show whether they are spatially clustered or randomly distributed.

2.7.2.5. Test D: normalization sensitivity (TOPSIS). Normalisation sensitivity is used to assess TOPSIS stability across 9 combinations of three normalisation approaches (Vector, Min-Max, Max) and three distance measures (Euclidean, Manhattan, Chebyshev; Çelen, 2014). A 9×9 correlation matrix is used to measure the concordance among pre-processing combinations. A close correlation ($\rho > 0.95$) indicates insensitivity to the methodology.

2.7.2.6. Test E: preference function analysis (PROMETHEE II). Preference functional sensitivity analyses PROMETHEE II stability on six types of functions: Type I (Usual), Type II (U-shape), Type III (V-shape), Type IV (Level), Type V (V-shape with indifference, baseline), and Type VI (Gaussian; Brans and Vincke, 1985). A 6-by-6 correlation matrix measures agreement between preference modelling methods. The correlations obtained are then evaluated to determine whether the baseline Type V preference function generates rankings identical to those of other preference structures.

2.8. Advanced validation techniques

2.8.1. Concordance prediction

ROC curve analysis (AUC) assessed rank difference accuracy in predicting categorical concordance, using category agreement as ground truth and absolute rank difference as predictor. At each threshold, sensitivity (true positive rate) and specificity ($1 - \text{false positive rate}$) were determined, and overall discriminative ability was summarized by the Area Under the Curve (AUC), where 0.7–0.8 indicates acceptable, 0.8–0.9 excellent, and >0.9 outstanding performance (Fawcett, 2006).

Bootstrap resampling was used to determine the stability and uncertainty of ROC-based concordance prediction, as has been historically the case in environment modeling (Islam et al., 2025). The entire number of bootstrap iterations (random_state = 42) was 100, which resulted in 95 percent confidence intervals of AUC estimates. This was beyond the recommended limits (Efron and Tibshirani, 1994), with small confidence (width = 0.05), small standard error (0.012) and convergence ($n = 1000$).

The maximization of sensitivity and specificity led to the best value of the classification based on the Youden Index ($J = \text{TPR} - \text{FPR}$). The classification outcomes (true positives, true negatives, false positives, false negatives) were summarized by a confusion matrix. The common measures of model performance were accuracy, precision, sensitivity, specificity and F1-score.

2.8.2. Indicator-level concordance network analysis

Indicator-level concordance assesses whether methodological disagreement is concentrated in specific indicators or distributed uniformly. For all 56 indicators, Spearman's ρ was calculated separately for TOPSIS and PROMETHEE II rankings, with concordance measured as:

$$\Delta\rho = |\rho_{\text{TOPSIS}} - \rho_{\text{PROMETHEE}}| \quad (7)$$

Smaller values signify higher concordance, where methods assign similar importance to indicators. Indicators were categorized as Very High ($\Delta\rho < 0.1$), High ($0.1 \leq \Delta\rho < 0.2$), Moderate ($0.2 \leq \Delta\rho < 0.3$), Low ($0.3 \leq \Delta\rho < 0.4$), or No Concordance ($\Delta\rho \geq 0.4$).

A network representation was built using NetworkX 3.1, with node sizes proportional to average correlation strength and colors denoting concordance class. Edges connect indicators with similar patterns ($|\Delta\rho_i - \Delta\rho_j| < 0.15$). The degree of connectivity is measured by network density (the ratio of actual to possible edges), where a value near 1.0 indicates a uniform distribution of concordance.

2.8.3. Spatial network analysis of geographic concordance patterns

Spatial network analysis of methodological concordance tests the implications of the spatial structuring of hydrological geography on methodological concordance. Nodes represent 1000 evaluation points, with edges connecting locations within 5 km to capture spatial correlations while preventing over-connection. The nodes were rated by ranking agreement: Strong Concordance ($\Delta\text{rank} \leq 50$), Moderate (51–100), Mild Discordance (101–200), and Strong Discordance (>200). The weight of an edge between two connected locations was the absolute difference in rank differences. Minimum Euclidean distances to major rivers (Brahmaputra, Teesta, and tributaries) were calculated using Geopandas 0.14.0 and Bangladesh Water Development Board River network geometries (bgd_watcrsa_1m_rivert.shp). Local connectivity (network degree) and river proximity relationships were quantified using node connections and Spearman's rank correlation. The binned locations were based on distance: 0–1 km, 1–2 km, 2–3 km, 3–4 km, 4–5 km, 5–10 km, 10–15 km, and >15 km. The statistical significance of differences between the degrees was tested with the Mann–Whitney U test using a Bonferroni correction ($\alpha = 0.05/7 = 0.007$) (Mann and Whitney, 1947; Bonferroni, 1936). River buffer zones (1 km, 2 km, 3 km) classified network edges as within-buffer, mixed, or outside-buffer based on endpoint proximity. Connectivity concentration is measured by edge density ratios (connections per node near rivers / connections per node far) and investigated using chi-square statistics ($\alpha = 0.05$).

2.9. Concordance and discordance hotspots analysis

To examine whether methodological discordance exhibits spatial clustering, spatial autocorrelation analysis was conducted using Moran's I and LISA (Anselin, 1995).

Global spatial autocorrelation was quantified using Moran's I statistic (Anselin, 1995), which measures the degree of spatial clustering in rank differences. For this study, Moran's I was calculated as:

$$I = \frac{N}{W} \times \frac{\sum_{i=1}^N \sum_{j=1}^N w_{ij} (RD_i - \overline{RD})(RD_j - \overline{RD})}{\sum_{i=1}^N \sum_{j=1}^N w_{ij} (RD_i - \overline{RD})^2} \quad (8)$$

where N is the total number of locations (1,000), RD_i is the absolute rank difference at location i ($|\text{TOPSIS rank} - \text{PROMETHEE rank}|$), RD_j is the rank difference at neighboring location j , $\overline{RD}$ is the mean rank difference, w_{ij} are spatial weights, and $W = \sum_{i=1}^N \sum_{j=1}^N w_{ij}$.

The spatial weight matrix was constructed using K-nearest neighbors (KNN) with $k = 8$, such that each location was connected to its eight nearest neighbors. This choice follows established spatial autocorrelation practice (Anselin, 1995) and balances local neighborhood representation with computational efficiency. Sensitivity analysis across $k = 6 - 10$ confirmed stable Moran's I estimate. Weights were row-standardized so that $\sum_j w_{ij} = 1$ for each location i .

Under the null hypothesis of spatial randomness, the expected value of Moran's I is $E[I] = -1/(N-1) \approx -0.001$ for $N = 1,000$. Statistical significance was assessed using 999 random permutations, with p

-values computed as the proportion of permuted statistics more extreme than the observed value.

LISA was used to identify local spatial autocorrelation, including clusters (High-High, Low-Low) and spatial outliers (High-Low, Low-High), at $\alpha = 0.05$. To complement rank-based analysis, categorical concordance was examined using resilience classes (Very Low, Low, Moderate, High, Very High) derived from AHP-TOPSIS and AHP-PROMETHEE II. Locations assigned to the same resilience category by both methods were classified as concordant; otherwise, they were classified as discordant. The spatial distribution of concordant and discordant locations was visualized to identify geographic patterns.

2.10. Statistical software and reproducibility

All analyses performed in Python 3.12.7 using NumPy 1.24.3, Pandas 2.0.3, SciPy 1.11.4, scikit-learn 1.3.2, NetworkX 3.1, geopandas 0.14.0, and PySAL 2.8.0. Spatial visualizations employed ArcGIS 10.8 (WGS 1984 UTM Zone 46 N). Complete supplementary materials include indicator definitions (Table S1), Saaty's scale (Table S2), union list (Table S3), weighted matrices (Table S4), AHP matrices with consistency (Table S5), descriptive statistics (Table S6), PROMETHEE thresholds (Table S7), weighting robustness (Table S8), MABAC/SPOTIS robustness (Tables S9–S10), and sensitivity results (Figures S1–S8). All analyses used fixed random seeds (random_state=42) for reproducibility.

3. Results

3.1. CFRI-Based flood resilience assessment

AHP-TOPSIS resilience scores (Fig. 3a) range from 0.278 to 0.569 (mean = 0.422, SD = 0.055), showing a significant spatial pattern in Kurigram District. The five-class grouping and Z-score standardisation indicate that the most frequent level is moderate resilience ($-0.5 \leq Z < 0.5$), with 36.1 per cent of locations (361/1000). High ($0.5 \leq Z < 1.5$) and very high ($Z > 1.5$) resilience are 24.6% (246) and 6.7% (67), respectively, and make up 31.3% of the district with above-average resilience. On the other hand, 25.8% (258) and 6.8% (68) of the locations are in the Low ($-1.5 \leq Z < -0.5$) and Very Low ($Z < -1.5$) resilience categories, respectively, accounting for 32.6% of the locations with below-average capacity. Both MCDA approaches exhibit similar spatial patterns, with higher resilience in central and southern unions and lower resilience in northern and western zones.

Fig. 3a shows that spatial analysis reveals significant geographic structuring. The central unions around Kurigram Sadar (e.g., 10, 11, 13–15) are characterised by the dominance of high to very high resilience due to the urban concentration of infrastructures, better access to institutions, and more diversified economic opportunities. The unions in the south (31–35) are more resilient because they are better connected to the market, have comparatively stable hydrology, and are not in active river channels. Conversely, the few northern border unions (16, 17, 21, 23, 24, 28, and 29) and western unions (39, 43, 46, and 49) also have dominant low-to-moderate resilience due to the dynamics of transboundary rivers, susceptibility to embankment and char land instability, riverbank erosion, and inadequate infrastructure development.

Frequency distributions of the six dimensions (Supplementary Fig. S1) show roughly normal distributions of TOPSIS scores, except for the Economic (mean = 0.45) and Infrastructural-Physical (mean = 0.48) dimensions, which exhibit slight right skewness due to the presence of high-capacity urban outliers. Categorical classification means that there is broadly balanced resiliency across dimensions: Socio-Cultural (35.5% Moderate; 6.7% High), Economic (34.7% Moderate; 4.9% High), Infrastructural-Physical (37.6% Moderate; 5.1% High), Organizational-Institutional (41.5% Moderate; 6.3% High), Hydraulic (41.8% Moderate; 7.4% High), and Ecological (41.8% Moderate; 7.2% High).

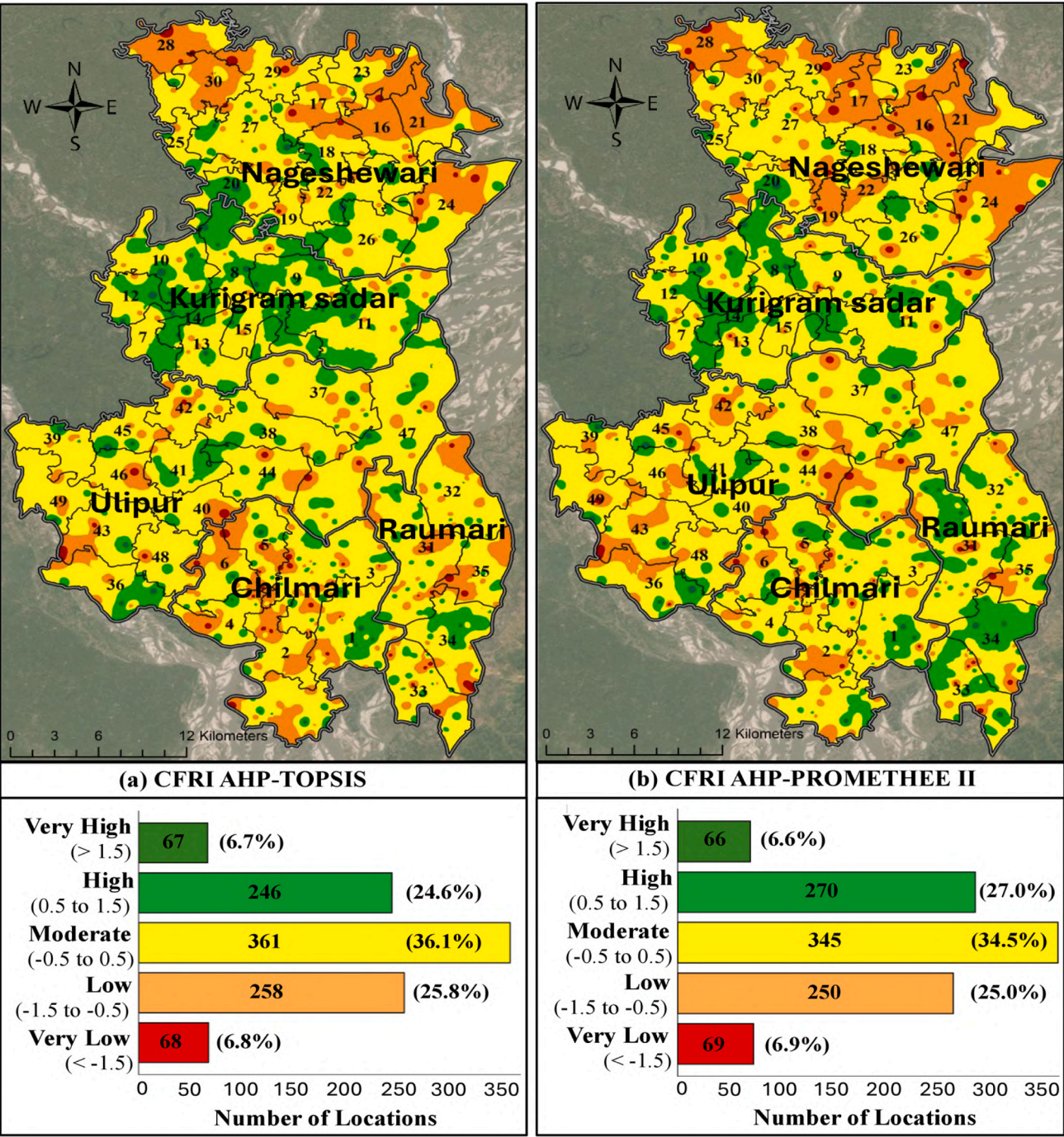


Fig. 3. Spatial distribution of flood resilience in Kurigram District using (a) AHP-TOPSIS and (b) AHP-PROMETHEE II methods with 56 indicators across 1000 locations. Categorical concordance: 62.0%.

Dimensional patterns (Supplementary Fig. S2) show socio-cultural and economic resilience concentrated in major urban centers; infrastructural-physical resilience with the greatest urban concentration; hydraulic resilience with a preference for engineered central areas; and ecological resilience with riparian vegetation and soil quality variability. Descriptive statistics (Supplementary Table S6) verify adequate differentiation ($CV = 12.98\%$; $range = 0.29$) and near-normal distribution ($skewness = 0.23$; $kurtosis = 2.87$), justifying Z-score standardization.

The scores on the PROMETHEE II net flow (Fig. 3b) are between -0.347 and 0.303 ($mean = 0.000$, $SD = 0.122$), which is the anticipated symmetric zero-centered distribution. Top scores in Z-score

classification are similar to TOPSIS: 34.5% Moderate, 27.0% High, 6.6% Very High, 25.0% Low, and 6.9% Very Low. Geographic concordance of methods is strongly concordant with central and southern unions of high-to-very-high resilience and with the northern border and western charlands unions of low-to-moderate resilience. All told, 62.0 per cent of locations have the same types of resilience, with the rest (38.0 per cent) showing shifts in single categories. The absence of spatial disagreement effects (Global Moran's $I = -0.0099$, $p = 0.5541$) confirms discordance stems from methodological aggregation differences rather than geography. Dimensional PROMETHEE II findings match TOPSIS (Spearman's $\rho = 0.84-0.91$, $p < 0.001$), with preference function testing validating strong Type V threshold performance ($\rho > 0.96$).

3.2. Tier 1 concordance analysis: correlation and agreement

3.2.1. Rank correlation and distribution analysis

Rank correlation analysis (Table 2, Part A) reveals very strong positive relationships between AHP-TOPSIS and AHP-PROMETHEE II rankings across all metrics. Spearman correlation ($\rho = 0.8741$, $p < 0.001$) indicates very strong monotonic agreement, and Pearson correlation ($r = 0.8799$, $r^2 = 0.774$) indicates that approximately 77% of the score variance is shared between methods. This result is independently validated by Kendall's tau ($\tau = 0.6912$, $p < 0.001$), which indicates that 69.1% of all location pairs exhibit consistent ordering. Consistently high statistical significance ($p < 0.001$) indicates that observed concordance stems from actual methodological agreement rather than sampling effects. On the whole, the magnitude of the correlation is significantly higher than that of traditional indicators of a strong association, indicating a high level of agreement even with an entirely different aggregation logic.

Rank difference analysis (Table 2, Part B) shows high correlation alongside significant positional displacement. The mean rank difference is 112.3 positions (11.2% of the ranking range), and the median of the rank difference is 92.0 positions, which is a right-skewed distribution. Although the changes in ranking are moderate in most locations, some locations show extreme differences, with 464 positions being the maximum. The standard deviation of 94.8 positions indicates significant variation between locations. The percentile analysis shows that 50% of locations differ by ≤ 92 ranks, 75% by ≤ 168 ranks, and 95% by ≤ 295 ranks. Fine-scale ranking prioritization requires caution; while extreme discordance (>300 ranks) is rare, moderate discordance (100–200 ranks) occurs in approximately 30% of locations.

3.2.2. Categorical concordance patterns

Categorical concordance analysis (Table 2, Part C) shows 52.9% of locations (529/1000) exhibit rank differences ≤ 100 , representing Strong (30.9%) or Moderate (22.0%) concordance. Strong Concordances (difference in rank < 50) indicate almost equal measures of the two techniques and high-confidence decision-making. The moderate level of Concordance (ranks 51–100) suggests acceptable correspondence, with slight changes in ranks not significantly altering the interpretation of resilience.

In contrast, 47.1% of locations (471/1000) show discordance exceeding 100 ranks, classified as Mild Discordance (101–200 ranks;

Table 2

Rank correlation coefficients, rank differences, and categorical agreement between AHP-TOPSIS and AHP-PROMETHEE II methods across 1000 locations in Kurigram District.

Metric	Value	p-value	Interpretation
A. Rank Correlation			
Spearman's ρ	0.8741***	$p < 0.001$	Strong positive correlation
Kendall's τ	0.6912***	$p < 0.001$	Substantial agreement
Pearson's r	0.8799***	$p < 0.001$	Strong linear relationship
B. Rank Differences			
Mean	112.3		Average positional shift
Median	92.0		Typical displacement
Maximum	464		Extreme case
C. Agreement Categories			
Strong Concordance (<50)	309 (30.9%)		High agreement
Moderate Concordance (50–100)	220 (22.0%)		Acceptable alignment
Mild Discordance (100–200)	300 (30.0%)		Moderate disagreement
Strong Discordance (>200)	171 (17.1%)		Methodological sensitivity

*** Highly statistically significant ($p < 0.001$).

300 locations, 30.0%) or Strong Discordance (>200 ranks; 171 locations, 17.1%). Mild Discordance can change intervention priorities but has little effect on the overall resilience subdivision. Strong Discordance describes an extreme situation of substantive divergence in methods that require reconciliation through further analysis or professional opinion. The proportion of Strong Discordance cases is relatively low in numbers, which proves that extreme rank reversals are sporadic and not regular tendencies.

Resilience category agreement confirms methodological robustness, with 62.0% of locations sharing identical classifications—significantly higher than the 20% expected by chance across five categories. Of the remaining 38.0%, 85% are single-category shifts, suggesting that most categorical disagreements are minor.

These results are supported by integrated visual diagnostics (Supplementary Fig. S5). Scatter plots demonstrate high linear relationships with low dispersion, rank difference distributions are skewed to the right (mean = 112.3; median = 92.0), confusion matrices reveal that the dominant relationships are on the diagonal and a Bland-Altman test indicates that discordance is place-dependent because it is not dependent on methods or magnitude.

3.3. Tier 2 robustness validation: stability and sensitivity analysis

3.3.1. Rank reversal stability analysis

Rank reversal analysis (Table 3) shows that PROMETHEE II exhibits substantially greater ranking stability than TOPSIS. The means of reversals in PROMETHEE II is 0.66 per cent, as against 3.23 per cent reversal in TOPSIS—a reduction of about 80 per cent. PROMETHEE II has very low reversal rates, regardless of the removal categorised as Top 10%, Bottom 10%, or Middle 10%; 0.87% for Top 10%, 0.89% for Bottom 10%, and 0.21% for Middle 10%.

TOPSIS, on the other hand, shows high scenario-specific instability, with reversal rates of 2.45% (top 10%), 4.31% (bottom 10%), and 2.93% (middle 10%). The greatest instability is in the withdrawal of low-performing options, which disproportionately affects TOPSIS rankings. The sensitivity of this approach stems from the distance-based nature of TOPSIS, where ideal and anti-ideal solutions are defined at the extremes of a dataset; eliminating the bottom-ranked options recalibrates the anti-ideal reference point, and rank adjustments are made for the remaining options.

PROMETHEE II demonstrates scenario-independent stability (0.21–0.89%), as its pairwise outranking framework depends on relative dominance relationships that persist regardless of alternative removal. The Spearman rank correlations support these results: PROMETHEE II maintains near-perfect rank order in all situations ($\rho = 0.9996$ – 0.9999 , $p < 0.001$), and TOPSIS shows slightly lower but still strong results ($\rho = 0.9901$ – 0.9966), with the lowest preservation observed when removing Bottom 10% situations.

3.3.2. Weight sensitivity analysis (OAT method)

Weight sensitivity analysis (Table 4) reveals stark methodological differences between TOPSIS and PROMETHEE II. TOPSIS is highly sensitive to weight changes, and the average Spearman correlation across the 56 indicators is 0.0491 ± 0.0035 , indicating that the system is very prone to changes in rankings when weights change by up to $\pm 30\%$. None of the indicators had a classification of Robust ($\rho \geq 0.95$) or Moderate ($0.90 \leq \rho < 0.95$); all 56 indicators (100 per cent) were classified as Highly Sensitive ($\rho < 0.90$), and the highest correlation was 0.0516.

In contrast, PROMETHEE II demonstrates exceptional robustness, with a mean correlation of 0.9986 ± 0.0003 , indicating near-complete preservation of rankings despite weight perturbations. All 56 indicators (100%) were classified as Robust ($\rho \geq 0.95$) with the lowest correlation of 0.9996. The very small standard deviation is a strong confirmation of the consistent strength of all indicators, and there are no indications of dimension- or indicator-sensitive characteristics.

Table 3

Rank reversal stability analysis comparing TOPSIS and PROMETHEE II across three removal scenarios (Top 10%, Bottom 10%, Middle 10% of 1000 locations). PROMETHEE II shows superior stability (RR = 0.66%) versus TOPSIS (3.23%), with 80% lower reversal rate and consistent performance across all scenarios.

Scenario	Method	Rank Reversal Rate (%)	Num Reversals	Spearman's ρ	p-value
Top 10%	TOPSIS	2.448,152,268	9904	0.996,645,936	0
Top 10%	PROMETHEE	0.868,372,265	3513	0.999,536,559	0
Bottom 10%	TOPSIS	4.305,030,281	17,416	0.990,012,324	0
Bottom 10%	PROMETHEE	0.892,596,712	3611	0.999,519,752	0
Middle 10%	TOPSIS	2.931,899,642	11,861	0.994,579,212	0
Middle 10%	PROMETHEE	0.212,334,693	859	0.999,958,683	0

Table 4

Weight sensitivity analysis using OAT method with $\pm 30\%$ perturbations for all 56 indicators. Shows mean correlation, standard deviation, and robustness classification for TOPSIS and PROMETHEE II methods.

Method	Min Correlation	Max Correlation	Mean $\pm$ SD	Robust ($p > 0.95$)	Moderate (0.90–0.95)	Sensitive (< 0.90)
TOPSIS	0.0298	0.0516	0.0491 $\pm$ 0.0035	0	0	56
PROMETHEE	0.9986	0.9999	0.9996 $\pm$ 0.0003	56	0	0

Indicator-level sensitivity patterns (Supplementary Fig. S6) further illustrate this contrast. The TOPSIS sensitivity scores ($1 - \rho$) range from 0.948 to 0.950, indicating that all indicators are similarly high in terms of instability. However, the PROMETHEE II sensitivity range is 0.0001–0.0007, confirming the insignificant ranking dispersion. These findings reveal that the TOPSIS rankings are very sensitive to the accurate specification of weights, whereas PROMETHEE II is stable under significant weight uncertainty.

3.3.3. Weight uncertainty analysis (MCS)

To quantify the variability in ranking within the uncertainty of weight, MCS (Fig. 4) spreads out weight uncertainty by running 10,000 trials. The findings are in line with the OAT findings on a scale. The probability density distribution has a very high central tendency (mean = 0.02; median = 0.02) and the spread is symmetric, implying that there is approximately normal behavior of ranking outcomes under uncertainty. Most of the simulated rankings fall within the range of the 95% confidence interval ([0.01, 0.03]). Extreme clustering at the mean (25th

percentile = 0.01; 75th percentile = 0.02) indicates high ranking stability, which confirms the fact that baseline rankings take a perfect reflection of the entire uncertainty space. Without bimodality or a heavy tail, weight patterns cannot produce distinct structures of rank. The spread measured, relative to the theoretical maximum variability, has been found to be roughly 5%, hence proving that the uncertainty in stochastic weight has very little impact on ranking results.

3.3.4. Spatial autocorrelation analysis (Global and local moran's I)

The spatial autocorrelation analysis presented in Section 3.6.3 (Spatial Autocorrelation and Discordance Hotspots) confirms that methodological discordance exhibits no significant spatial clustering. Global Moran's $I = -0.0099$ (Expected $I = -0.0010$, $Z = -0.5917$, $p = 0.5541$) indicates that rank variations between TOPSIS and PROMETHEE II are not concentrated in geographic areas. The LISA analysis shows that 10.1 percent of locations are statistically significantly clustered ($p < 0.05$), and the rest (89.9%) show no spatial autocorrelation.

This observation supports the conclusion that methodological

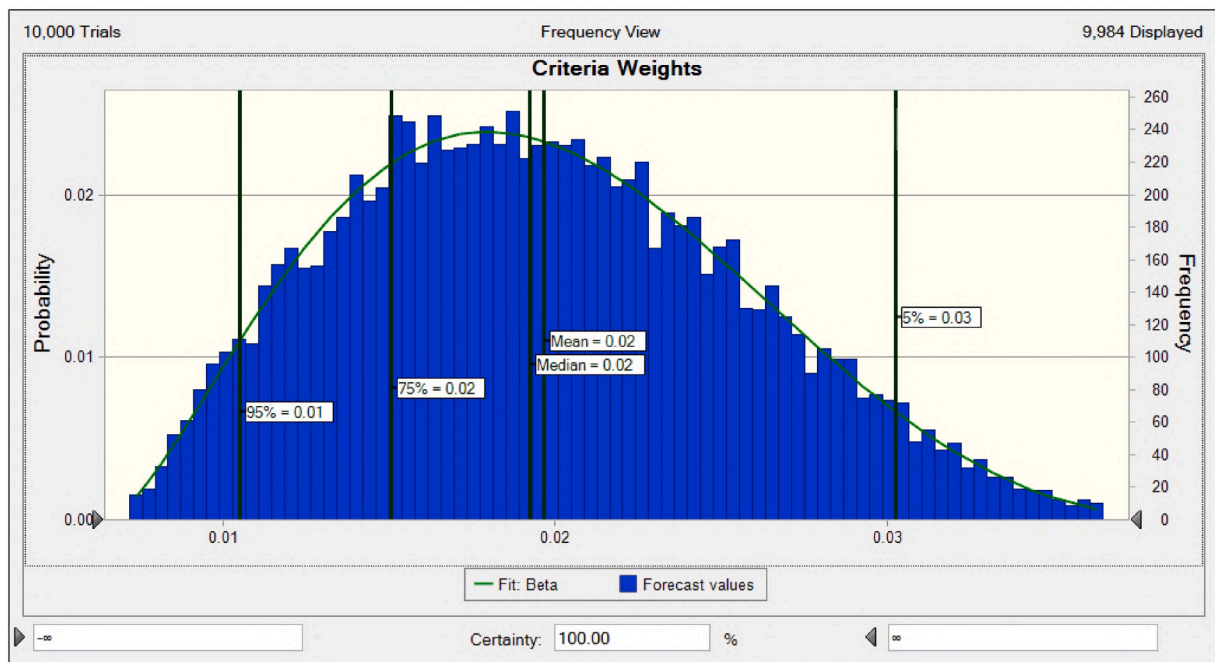


Fig. 4. MCS of weight uncertainty propagation through 10,000 trials showing probability density distribution of ranking variations. Central tendency (mean = 0.02, median = 0.02) with 95% confidence interval [0.01, 0.03] demonstrates ranking stability underweight perturbations.

discordance is not due to location-specific factors such as the gradient of flood exposure, socio-economic factors, or union-level systems of governance, but rather to inherent mathematical differences between distance-based (TOPSIS) and outranking (PROMETHEE II) aggregation approaches. The geographic randomness of discordance makes both approaches confident in their strength because nothing is systematic across geographic space that would make the models irrelevant to the context of various floodplains.

3.3.5. Normalization and preference function sensitivity

Normalization sensitivity analysis for TOPSIS (Supplementary Fig. S7) reveals that ranking outcomes depend substantially on pre-processing choices. The correlation matrices expected under the nine TOPSIS variants (3 normalizations by 3 distance measures) have correlation coefficients (ρ) ranging from 0.42 to 0.98 and reflect variable concordance across combinations of the methods used. The baseline vector normalization using Euclidean distance shows a high correlation with min-max normalization using Euclidean distance ($\rho = 0.94$), but a lower correlation with max normalization using Manhattan distance ($\rho = 0.58$), validating the idea that normalization-distance interactions do not bypass TOPSIS rankings in an otherwise trivial sense. Analysis of preference functions in PROMETHEE II (Supplementary Fig. S8)

indicates rare stability in all six types of preference functions. The 6×6 correlation matrices have high concordance levels (minimum $\rho = 0.96$, maximum $\rho = 1.00$, mean $\rho = 0.98$), indicating that PROMETHEE II rankings are insensitive to the choice of preference functions. Type V (V-shape with indifference, baseline) has very strong correlations with Type III (V-shape, $\rho = 0.99$) and Type IV (level, $\rho = 0.98$), and even the most differentiated function, Type I (usual), has a high concordance ($\rho = 0.96$).

3.4. Advanced validation: ROC-Based concordance prediction

ROC curve analysis (Fig. 5a) demonstrates that absolute rank difference exhibits strong discriminative ability for predicting categorical concordance between AHP-TOPSIS and AHP-PROMETHEE II, achieving an AUC of 0.793 (95% CI: [0.772, 0.822]). The given AUC value reflects acceptable to excellent discrimination per the traditional rules for interpreting this value, and it has been confirmed that rank difference is an effective proxy for categorical agreement. The ROC curve shifts significantly away from the diagonal (AUC = 0.5, random guessing), indicating that rank difference versus categorical concordance is a systematic association rather than a coincidental one.

The distribution of AUC across 100 resampling runs is stable, with a

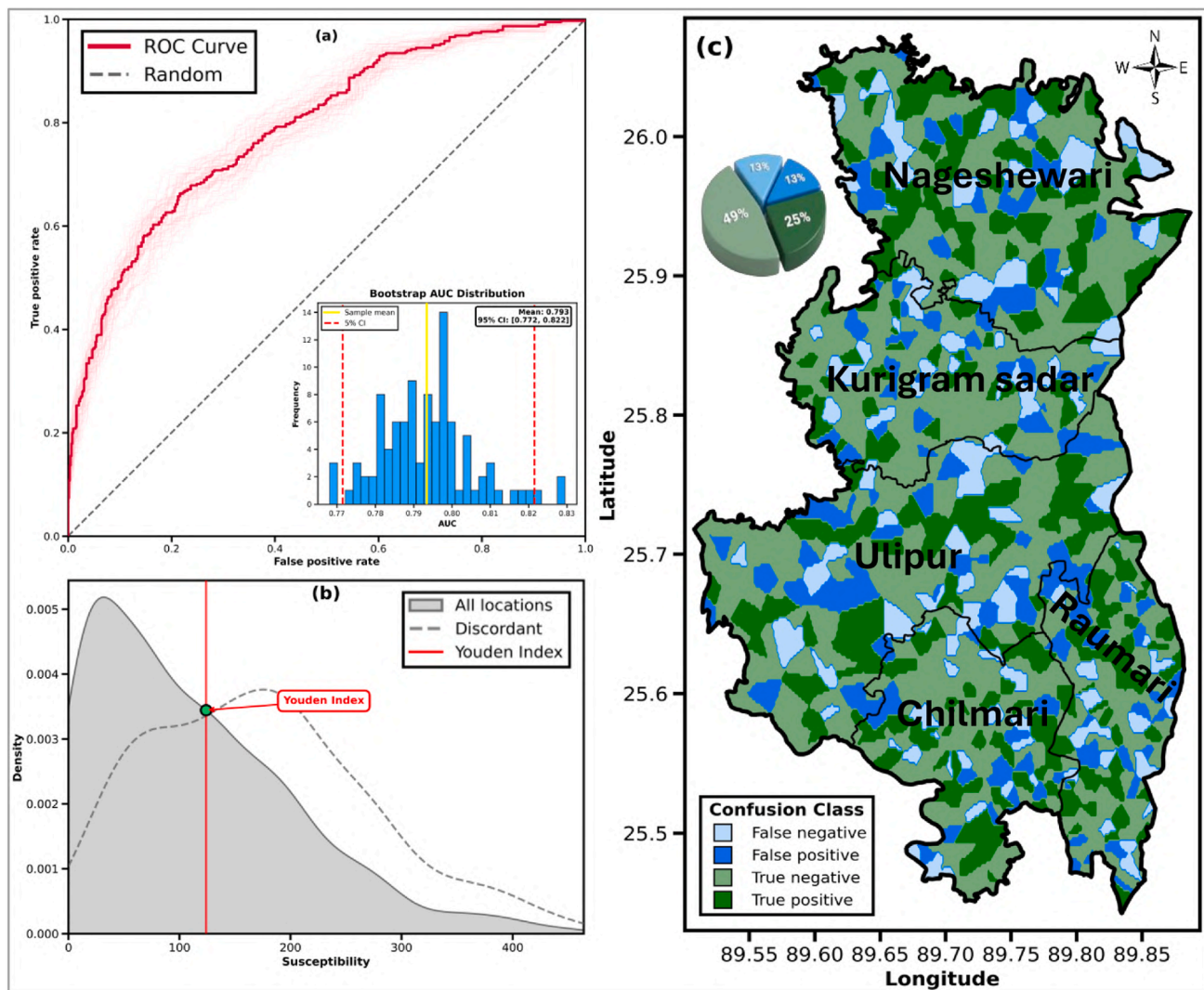


Fig. 5. ROC-based validation of concordance prediction using rank differences between TOPSIS and PROMETHEE II across 1000 locations. (a) ROC curve (AUC = 0.793, optimal threshold = 124 positions), (b) Youden Index identification, (c) Spatial confusion matrix showing True Positive (25.2%), True Negative (48.6%), False Positive (13.4%), and False Negative (12.8%) distributions.

mean AUC of 0.793 and a narrow 95% confidence interval (0.772 to 0.822), unaffected by sampling variation, indicating that the model is strong. The index of the Youden optimal classification threshold (Fig. 5b) is at a rank difference of 124, with a Youden index of $J = 0.447$. The sensitivity (true positive rate) and specificity (true negative rate) of the model at this threshold are 0.653 and 0.794, respectively, meaning that 65.3 per cent of categorical discordances are accurately predicted, whereas 79.4 per cent of categorical concordances are correctly identified. The optimum threshold of 124 ranks represents approximately 12.4% of the total possible ranks, so a convenient decision rule is that

locations differing by >124 ranks between methods are expected to fall into different resilience categories, while those differing by 124 ranks or less are expected to be in the same category.

The confusion matrix reveals an overall classification accuracy of 73.8% (738/1000 locations correctly classified), including 252 True Positives (25.2%), 486 True Negatives (48.6%), 134 False Positives (13.4%), and 128 False Negatives (12.8%). The precision of 0.653 implies that 65.3 per cent of predicted discordances are actual discordances, whereas the F1-score of 0.664 reflects a balanced performance in detecting both concordant and discordant locations. Geographic

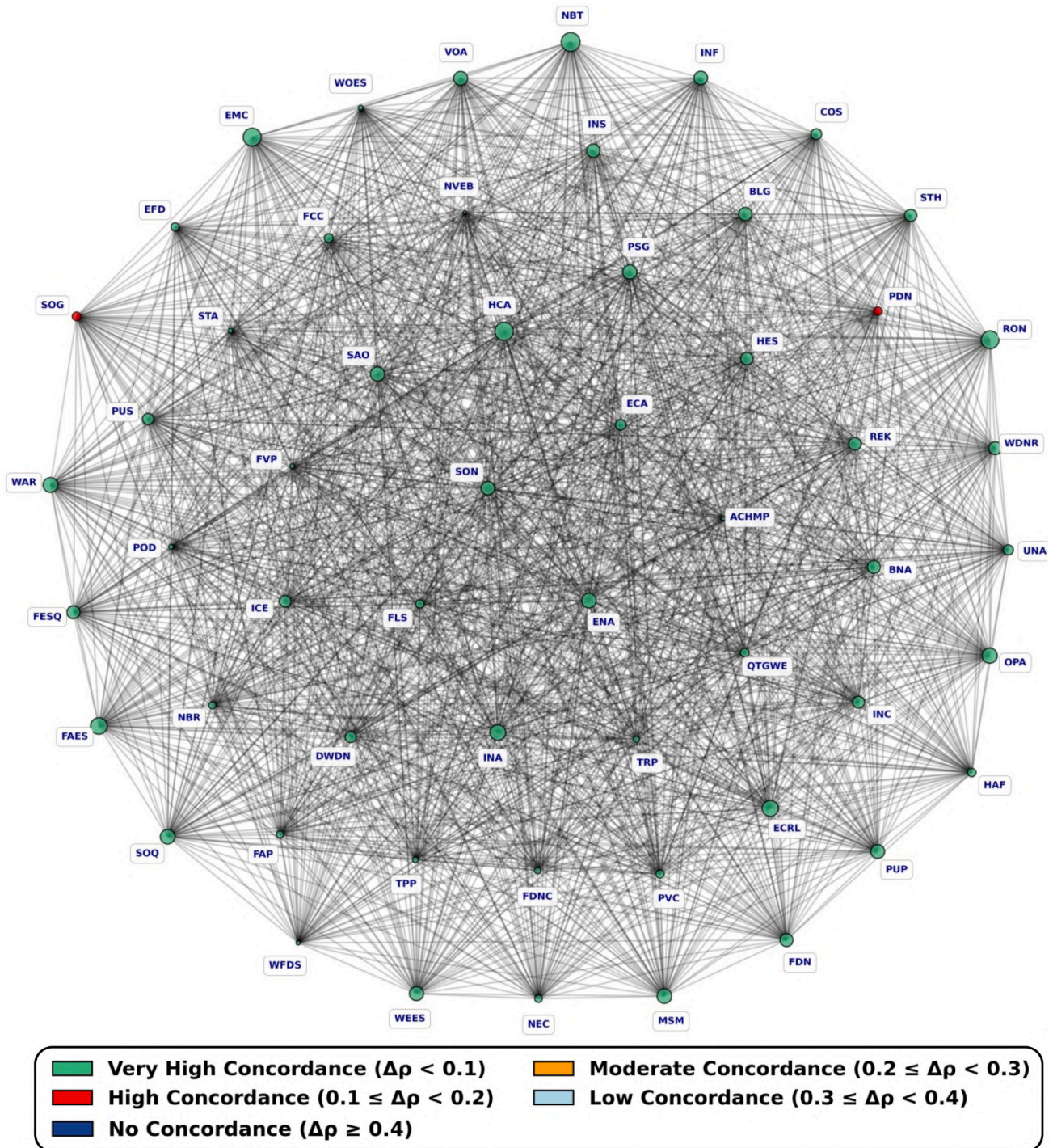


Fig. 6. Indicator-level concordance network between AHP-TOPSIS and AHP-PROMETHEE II across 56 indicators. Node size represents correlation strength; colors denote concordance levels based on $\Delta\rho = |\rho_{\text{TOPSIS}} - \rho_{\text{PROMETHEE}}|$: Very High (green, $\Delta\rho < 0.1$, 96.4%), High (red, $0.1 \leq \Delta\rho < 0.2$, 3.6%). Network density = 1.0 indicates complete interconnectedness.

patterns in the performance of classifications are revealed in the form of spatial distribution of prediction classes (Fig. 5c).

The northern areas of the floodplains of Nageshewari and some portions of Kurigram Sadar are also dominated by true negatives (light green, correctly predicted concordant locations), which constitute 48.6 percent of the study area. True Positives (dark green, correctly identified discordant locations) make up 25.2 percent of the locations and are found throughout the district, with a slight concentration in riverine transitional areas. False Positives (blue, falsely predicted discordant yet actually concordant) are distributed in 13.4 percent of the locations and occur in the middle-range resilience regions of Ulipur, Raumari and southern Chilmari. False negatives (light blue, predicted concordant when it was discordant) supply a 12.8% location rate and are scattered through all the subdistricts with no clear geographic focus.

3.5. Indicator-Level concordance network analysis

Indicator-level concordance analysis (Fig. 6) demonstrates highly consistent methodological agreement across the 56-indicator flood resilience framework. A total of 96.4% of indicators (54/56) exhibit very high concordance ($\Delta\rho < 0.1$), indicating that AHP-TOPSIS and AHP-

PROMETHEE II assign nearly identical relative importance to almost all criteria. Only Slope of Ground (SOG; $\Delta\rho = 0.1265$) and Population Density (PDN; $\Delta\rho = 0.1233$) show slightly lower high concordance. All indicators are above this level, confirming that no single criterion drives systematic methodological disagreement.

Complete connectivity (density = 1.0) is observed in the concordance network, indicating homogeneous agreement across all indicators, rather than clustering by indicator type (e.g., physical and socio-economic). The quality of influential indicators, including distance from rivers, road density, and healthcare accessibility, is characterized by strong correlations and high concordance, supporting their strength in decision-making frameworks. The presence of very high concordance node dominance allows us to state that methodological agreement is not ad hoc. All these findings suggest that differences in location-level rankings can be attributed to cumulative aggregation effects across indicators rather than to sensitivity to individual criteria, implying that the observed discordance is inherent to the nature of distance-based and outranking aggregation logics rather than to indicator selection bias.

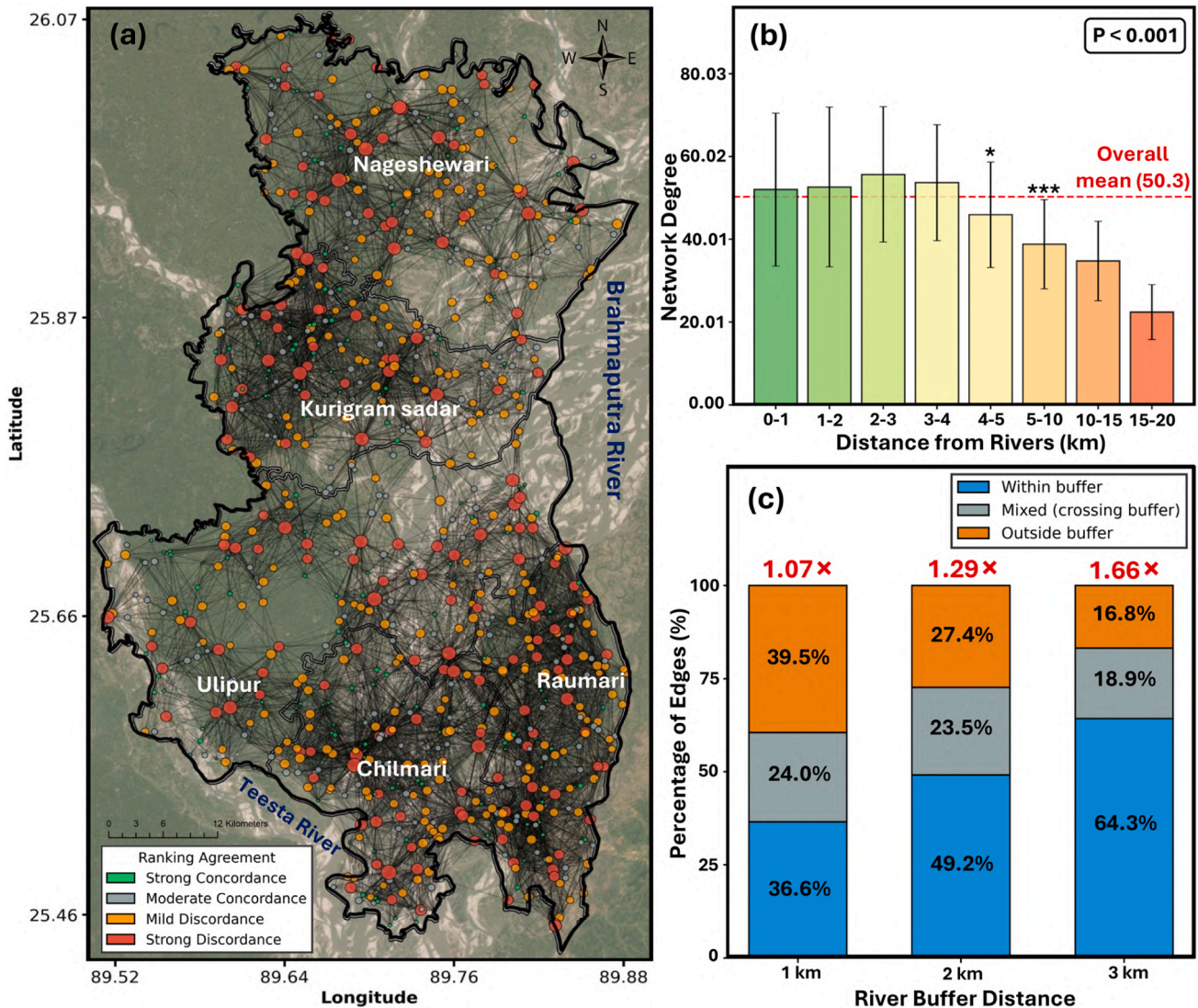


Fig. 7. Spatial network analysis of methodological concordance across 1000 locations. (a) Network visualization showing locations colored by agreement: Strong Concordance (green, 30.9%), Moderate Concordance (gray, 22.0%), Mild Discordance (orange, 30.0%), Strong Discordance (red, 17.1%). Edges ($n = 25,134$) connect locations within 5 km. (b) Distance-decay relationship: network connectivity declines 57.5% from riverine zones to inland areas, $\rho = -0.218$, $p < 0.001$. (c) Edge density by river buffer zones shows 64.3% of edges occur within 3 km riverine zones despite comprising only 58.1% of locations ($\chi^2 = 892.4$, $p < 0.001$).

3.6. Spatial resilience patterns: methodological concordance and discordance

3.6.1. Spatial network analysis

Spatial network analysis (Fig. 7a) reveals pronounced geographic structuring of methodological concordance across Kurigram District, with 25,134 spatial connections (edges) linking the 1000 assessment locations within the 5 km proximity threshold. Visual analysis of the network map shows that connections are dense along the Brahmaputra River valley (eastern border) and the Teesta River area (southwestern region), and sparse in the central areas of inland Kurigram Sadar. The color distribution shows spatial heterogeneity among patterns of agreement: Strong Concordance places (green, $n = 309$, 30.9%) and Moderate Concordance places (gray, $n = 220$, 22.0%) take up 52.9 percent of the study area, with Mild Discordance places (orange, $n = 300$, 30.0%) and Strong Discordance places (red, $n = 171$, 17.1%) taking up the remaining 47.1 percent. A significant negative spatial relationship between river proximity and network connectivity (Fig. 7b) supports the result (Spearman $\rho = -0.218$) showing that the degree of network is greater at the location near large rivers compared to the inland environment. The mean 1 km connectivity of locations with rivers ($n = 195$) is 52.7 ± 18.2 SD, and 22.4 ± 6.6 SD with locations further than 15 km of rivers ($n = 12$), which is a 57.5 per cent decline in connectivity with increasing distance to rivers. Fig. 7c is the edge density analysis, an analysis of progressive concentration of network connections around hydrological features. At the 1 km buffer, 36.6% of edges are in riverine zones, which rises to 49.2% at the 2 km and 64.3% at the 3 km, although riverine zones only make up 58.1% of the assessment locations at the 3 km buffer ($\chi^2 = 892.4$, $p = 0.001$). There is a progressive increase in the edge density ratios: 1.07 x at 1 km ($p < 0.001$), 1.29 x at 2 km ($p < 0.001$), 1.66 x at 3 km ($p < 0.001$) which

indicates that there is marked over-representation of network edges in flood-prone areas.

3.6.2. Spatial concordance patterns

The spatial distribution of absolute rank differences (Fig. 8a) reveals substantial geographic heterogeneity in methodological concordance across Kurigram District. The range of rank differences was 0 (perfect agreement) to 464 (maximum disagreement), with a mean of 112.28 and a median of 92.00, indicating a right-skewed distribution, with most locations showing moderate discordance. The analysis of magnitude-based distribution of rank differences (Table 2, Part C) indicates that about half of the study area (52.9%) has rank differences < 100 , of which 30.9% has strong concordance (< 50) and 22.0% has moderate concordance (50–100) and the others of the 47.1% have mild to strong discordance (100–200: 30.0%; > 200 : 17.1%). The gradient of color yellow indicates that the highest rank differences (> 250) are concentrated in the northern floodplain unions and selected riverine areas along the Brahmaputra corridor, and the central administrative areas that surround Kurigram Sadar have lower rank differences (< 75), which points to more methodological stability in those zones.

Locations with the highest rank differences spatially co-locate with the zones of highest network connectivity identified in Section 3.6.1, particularly within 3 km of major rivers; this pattern is interpreted further in Section 4.3.

It is interesting to note that the lack of large areas of extensive discordance suggests that the methodological disagreement is not localized in particular geographic subregions but has a scattered, place-specific distribution, reflecting local resilience and gradations in flood exposure.

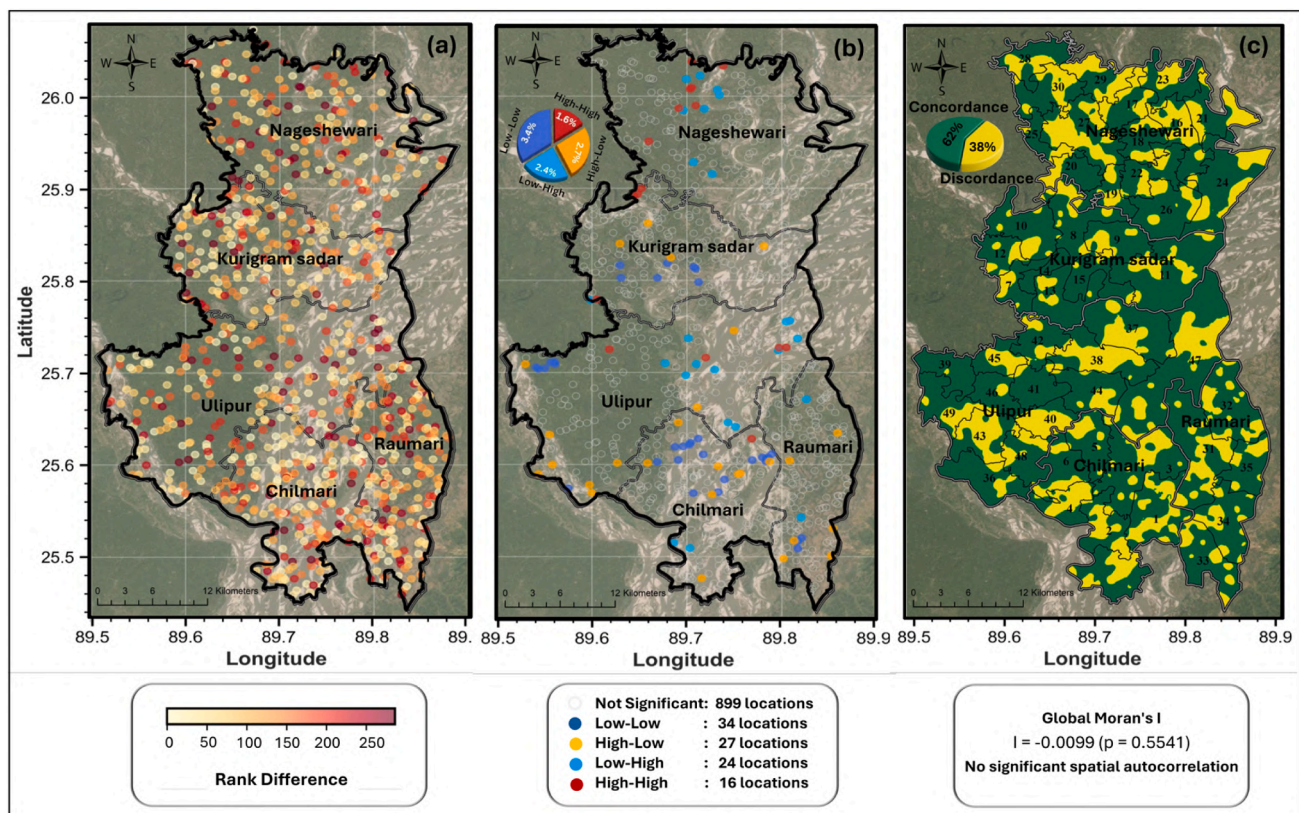


Fig. 8. Spatial analysis of methodological concordance and discordance between TOPSIS and PROMETHEE II rankings. (a) Rank difference distribution across 1000 locations (mean = 112.28, median = 92.00, range = 0–464). (b) LISA cluster map: High-High (red, 1.6%), Low-Low (blue, 3.4%), spatial outliers (orange/cyan, 5.1%), non-significant (gray, 89.9%). (c) Categorical concordance: exact match (teal, 62.0%) versus one-category difference (yellow, 38.0%). Global Moran's I = -0.0099 ($p = 0.5541$) confirms spatial randomness of discordance.

3.6.3. Spatial autocorrelation and discordance hotspots

Global Moran's I analysis ($I = -0.0099$, Expected $I = -0.0010$, Z -score $= -0.5917$, $p = 0.5541$) confirms no significant spatial autocorrelation in rank differences, indicating that methodological discordance appears spatially random rather than geographically structured (Figs. 8b, c). The observed Moran's I is statistically indistinguishable from the expected value under spatial randomness, and the non-significant p -value ($p > 0.05$) provides strong evidence that discordance is not driven by location-specific factors or regional clustering processes.

LISA cluster analysis (Fig. 8b) identifies only 101 locations (10.1%) exhibiting statistically significant spatial clustering ($p < 0.05$), with the remaining 899 locations (89.9%) classified as Not Significant. Among significant clusters, low-low clusters (blue, $n = 34$, 3.4%) represent concordance "cold spots" where focal locations and their neighbours both exhibit low rank differences, mostly found in central administrative units. The high-high clusters (red, $n = 16$, 1.6%) represent discordance "hot spots" concentrated in a few northern floodplain zones. Spatial outliers, including High-Low (orange, $n = 27$, 2.7%) and Low-High (cyan, $n = 24$, 2.4%), are spread across the district, indicating localised methodological sensitivities rather than systematic geographical features. The pie chart inset shows that non-significant locations are predominant, visually demonstrating that significant spatial autocorrelation is not present in any area.

The categorical concordance analysis (Fig. 8c) indicates that both methods classified 62.0% ($n = 620$) of the locations into the same resilience category, and 38.0% ($n = 380$) of the locations differed by one category. Concordant (teal) and discordant (yellow) areas are arranged in an indefinite pattern, and both types are mixed throughout the five sub-districts. This is in line with the I statistic of the Global Moran, which means that categorical agreement is randomly distributed spatially. The 62:38 concordance-discordance split shown in the pie chart inset is high, but with poor methodological agreement.

The topic of disagreement on space distribution confirms that the disagreement in approach is not contingent upon location-based factors, such as union-level governance, concentration of flood risk, or local socio-economic conditions. This finding adds to the confidence in the strength of both methods because neither is characterised by systematic geographic bias, which can restrict the use of either method in a broad spectrum of floodplain situations.

4. Discussion

4.1. Understanding methodological concordance and discordance

The comparative analysis shows high, albeit not perfect, concordance between AHP-TOPSIS and AHP-PROMETHEE II. The high monotonic agreement is also indicated by strong Spearman rank correlation ($\rho = 0.8741$, $p < 0.001$) and Pearson correlation ($r = 0.8799$, $r^2 = 0.774$), and by the fact that around 77% of the variance in resilience scores is shared. This finding indicates that there is overall consistency in the ranking results, even though TOPSIS and PROMETHEE II, have fundamentally different aggregation logics. Furthermore, the per cent categorical concordance was 62.0, which is significantly higher than the 20% level of agreement expected by random chance alone, and thus, reliable broad resilience stratification is found.

However, significant differences are revealed at the level of individual places. Such a high deviation in position ranking indicates a significant change in relative rank across the 112.3 mean positions (median = 92.0; maximum = 464). Places with rank variations over 200 places (i.e., Strong Discordance) and 17.1% of the sample should be reconciled through further sensitivity analysis or expert analysis, followed by prioritisation of intervention. Relevantly, among the 38.0 per cent of sites with categorical disagreement, about 85 per cent of the discordances involve single-category shifts rather than multi-category reversals, suggesting that the discordances are moderate rather than

extreme.

The fact that these differences are attributed to the indicator level further explains the level of concordance analysis. Most indicators (96.4%, 54 out of 56) show Very High Concordance ($\Delta\rho < 0.12$), with only the slope of ground (0.1265) and population density (0.1233) showing slightly lower agreement. This trend supports the idea that differences in ranking at the location level are due to cumulative and distributed differences across multiple indicators, driven by different aggregation strategies rather than sensitivity to a single indicator (Rizaliali et al., 2026).

While aggregate concordance metrics (Spearman $\rho = 0.8741$) indicate strong agreement, this should not be interpreted as a guarantee of consistent decision support at the individual location level: a mean rank displacement of 112.3 positions (11.2% of the ranking range) means that for a typical location, the two methods can disagree by more than one-tenth of the entire study area's rank order, and the maximum displacement of 464 positions represents a location ranked in the top half by one method and the bottom half by the other. For practical flood mitigation prioritisation, where resource allocation decisions for specific unions or households depend on rank position rather than aggregate correlation, this means method choice can materially change which locations receive priority intervention, particularly for the 17.1% of locations classified as strong discordance. We therefore recommend that rank-based prioritisation decisions in operational settings be supplemented by the ROC-based screening threshold (124 rank positions) to flag locations where methodological disagreement is large enough to warrant additional validation before resource allocation.

4.2. Robustness and stability under uncertainty

The results of the robustness analysis indicate that there are significant variations in the stability of AHP-TOPSIS and AHP-PROMETHEE II during the state of uncertainty, with a direct consequence to the assessment of flood resilience. Rank reversal testing shows that PROMETHEE II is much more stable with an average reversal rate of 0.66% compared to TOPSIS's reversal rate of 3.23%—an 80% decrease. The reversal rates of PROMETHEE II consistently remain very low (0.21–0.89%) because it is highly insensitive to different scenarios, whereas TOPSIS reaches a maximum reversal rate of 4.31% when the bottom 10% of alternatives are excluded. Topologically, TOPSIS reformulates optimal solutions using the limits of the data, growing rank variations, but the pairwise outranking associations in PROMETHEE II are always relative, irrespective of the data structure and therefore are much more stable. Near-perfect Spearman correlations of PROMETHEE II (0.9996–0.9999) demonstrate this stability benefit, in contrast to slightly smaller values of TOPSIS (0.9901–0.9966).

Weight sensitivity analysis reveals stronger divergence: at less than $\pm 30\%$ perturbations, TOPSIS exhibits extreme sensitivity ($\rho = 0.0491$), whereas PROMETHEE II is highly robust ($\rho = 0.9986$) with all indicators of robustness. This difference is essential in flood-resilience settings where the weights of indicators by experts are always uncertain. The fact that the indifference and preference thresholds in PROMETHEE II ($q = 0.20$, $p = 0.60$) weaken any effects of marginal performance differences makes it stable.

Monte Carlo simulations (10,000 trials) indicate the strength of PROMETHEE II also with narrow variability (95% CI: 0.01–0.03). This pattern is backed by sensitivity tests related to normalization and preference-function specification: TOPSIS rankings are highly variable ($\rho = 0.42 - 0.98$), whereas PROMETHEE II rankings are highly concordant ($\rho = 0.96 - 1.00$).

ROC analysis to operationalize results shows that absolute rank difference is a reliable predictor of categorical discordance (AUC = 0.793). The best cut-off of 124 rank positions produces a workable screening rule to determine the locations that need to be looked into before prioritizing interventions. The overall proportion of classification is 73.8%, which means that it is able to make predictions with certainty.

All these results point to the fact that both TOPSIS and PROMETHEE II can provide similar overall rankings, but they differ markedly in their robustness characteristics under the specific stability tests applied here: rank reversal (0.66% vs. 3.23%), weight sensitivity ($\rho = 0.9986$ vs. 0.0491), and normalization/preference-function sensitivity (Section 3.3.5). On the basis of these robustness criteria alone, PROMETHEE II would be the more defensible choice in decision contexts where indicator weights are uncertain or contested and ranking stability is operationally critical; however, this recommendation reflects robustness performance under the perturbation scenarios tested here, not a validated comparison against observed flood outcomes, and should be weighed against TOPSIS's computational simplicity and interpretability where weights are well-established and rapid, exploratory screening is the priority.

4.3. Spatial patterns of methodological concordance and implications for flood risk management

Spatial analysis reveals marked geographic structuring in methodological interaction frequency between AHP-TOPSIS and AHP-PROMETHEE II yet confirms the absence of systematic geographic bias. A network analysis was performed to reveal 25,134 spatial relationships within a 5 km buffer, revealing that the highest densities of these connections were concentrated along the Brahmaputra and Teesta River channels. Network connectivity is significantly greater in riverine settings; edge density is ~ 1.66 times higher within 3 km of major channels. The resultant distance-decay correlation (Spearman $\rho = -0.218$, $p < 0.001$) suggests that interactions as methodological variables increase with the hydrological approach, driven by the complexity and variability of flood indicators, which are much more apparent in active floodplain settings.

Although there is an increase in the rate of this interaction, the proximity to a river is not statistically converted into a more methodologically severe disagreement. Strong Discordance (rank differences of >200 positions) is only a little more likely to occur in riverine locations (12.3%) than in inland (10.8%), but the difference is not significant. While flood-exposed locations are ranked more frequently across methods, the magnitude of disagreement remains consistent across all geographical contexts. That is, riverine complexity makes differences more likely to be ranked, but does not necessarily lead to instability in the methodological situation.

Global spatial autocorrelation analysis confirms spatially random rank differences ($I = -0.0099$, $p = 0.5541$), indicating no systematic geographic bias. Unlike concentrated disagreement, which would suggest environment-specific advantages for either method, this randomness supports the broad applicability of both TOPSIS and PROMETHEE II across heterogeneous floodplain landscapes.

LISA analysis shows significant clustering at only 10.1% of locations; nearly 90% lack spatial structure. Low-Low types of concordance are clustered in the central administrative union, where both techniques are always consistent in assigning the level of resilience, but High-High types of discordance clusters are found intermittently in the northern floodplain regions. The fact that these clusters are both limited and widely spread suggests that methodological controversy is largely due to inherent differences in the logic of aggregation rather than to local differences in socio-economic, infrastructural, or hydrological conditions.

Categorical concordance mapping provides visual confirmation through complementation. Approximately 62% of locations yield identical resilience results under both approaches, while the remaining 38% differ by one category. These disparities are distributed across the study area with no observable geographic pattern. A combination of these findings suggests that methodological divergence is space-neutral and driven by mathematical aggregation properties rather than spatial context.

These spatial findings yield significant implications for flood risk

management. First, the lack of systematic geographic bias permits the use of either AHP-TOPSIS or AHP-PROMETHEE II across any landscape without risk of spatial setting introducing causal bias. Second, sites with Strong Discordance (17.1%) require priority validation, stakeholder consultation, or sensitivity analysis to ensure prioritisation remains method-independent; our ROC-based model ($AUC = 0.793$) provides an effective screening tool for such areas. Third, while high categorical concordance supports either approach for broad resilience stratification, fine-scale rank-based prioritisation remains sensitive to methodological logic and requires explicit documentation. Finally, the results highlight the importance of robustness in participatory settings: PROMETHEE II's stability underweight uncertainty is ideal for stakeholder-driven weighting, whereas TOPSIS's sensitivity necessitates high precision in weight specification. Overall, spatial analysis confirms that methodological agreement is driven by aggregation logic rather than geography.

The hydraulic and ecological dimensions warrant explicit ecohydrological interpretation, as their constituent indicators capture processes that are mechanistically, not just statistically, linked to flood resilience. Within the hydraulic dimension, drainage network indicators (Flood Drainage Network, Clean Drainage Network, Drainage Performance in Rainfall, Disposal of Waste in Drains, and Overall Drainage Performance) jointly describe the capacity of constructed and natural conveyance systems to evacuate floodwater, while Unpaved Areas and Steep Areas indicators capture infiltration capacity and overland flow generation, both first-order controls on local flood hydrology independent of socio-economic capacity. Notably, the Well-Functioning Drainage System indicator within the ecological dimension explicitly captures drainage designed in alignment with natural hydrology rather than purely engineered conveyance (Fletcher et al., 2015), conceptually bridging the hydraulic and ecological dimensions. More broadly within the ecological dimension, biodiversity/natural area and habitat fragmentation indicators reflect the degree to which wetlands, forests, and other natural habitats retain their flow-attenuation and water-regulation functions (Millennium Ecosystem Assessment, 2005; Folke, 2006; Foley et al., 2005), while the anthropogenic pressure indicator captures human-induced degradation, deforestation and encroachment (IPCC, 2014) that erodes these same ecological buffers. The concentration of strong discordance locations along the northern floodplain and riverine corridor (Section 3.6.3) coincides spatially with the zones where these hydraulic and ecological processes are most dynamic, including active channel migration, char-land instability, and drainage network disruption, suggesting that methodological disagreement between TOPSIS and PROMETHEE II is partly a signature of genuine underlying ecohydrological volatility rather than purely a mathematical artifact of aggregation logic.

4.4. Methodological contributions, advantages, and limitations

4.4.1. Methodological contributions

The work introduces the first high-resolution comparison of AHP-TOPSIS and AHP-PROMETHEE II to multidimensional flood resilience, using 1000 spatial units and 56 indicators on six dimensions in the largest delta system in the world (Raff et al., 2023).

Existing comparative MCDA work in flood risk falls into two categories, neither of which integrates weighting-method robustness, aggregation-method robustness, and spatial validation simultaneously. The first compares weighting methods within a single aggregation framework: Esmaili and Karipour (2024) compared subjective and objective weighting schemes for a flood hazard index but did not compare alternative ranking/aggregation methods or assess spatial structure of disagreement; Taoukidou et al. (2025) similarly compared AHP, Fuzzy AHP, and Frequency Ratio weighting but applied a single aggregation logic throughout. The second category applies a single MCDA method without cross-method validation: Platiridis and Mallios (2024) mapped flood susceptibility using PROMETHEE alone, and Rizaliali et al. (2026) and Aroca-Jiménez et al. (2023) each developed

a single hybrid or composite index without comparing it against an alternative aggregation logic. The present study differs by combining (i) two structurally distinct aggregation methods (distance-based and outranking-based), (ii) both subjective and objective weighting schemes (Section 2.5, Supplementary Table S8), and (iii) explicit spatial and statistical robustness validation across 1000 locations and 56 indicators, providing convergent evidence on method choice that none of the cited studies independently offer.

Combining correlation measures, rank-reversal tests, weight sensitivity, MCS, and ROC-based discrimination, the multi-layer scheme outperforms traditional MCDA comparisons to offer convergent evidence of concordance and strength.

Another contribution is the explicit spatial analysis of methodological agreement. The study introduces a spatial network structure and LISA clustering to the discordance ranking, showing that ranking is not geographically clustered—a feature that is seldom studied in MCDA validation.

Lastly, the rank-difference threshold based on ROC (124 positions; $J = 0.447$) provides a realistic decision rule to evaluate where there is a potential that methodological disagreement could influence prioritization of intervention.

4.4.2. Advantages of the methodological approach

Both AHP-TOPSIS and AHP-PROMETHEE II are transparent, deterministic and totally auditable, permitting the explicit interpretation of the impact that indicator values and weights will have on rankings. Such transparency is in contrast to the data-driven black-box models and promotes accountability in governance of flood risks. Combining these 56 indicators that cover social, economic, institutional, infrastructural, environmental, and hydraulic dimensions allow a holistic expression of flood resilience in line with the Sendai Framework (UNDRR, 2015) and EU Floods Directive (European Parliament and Council, 2007). This multidimensional integration is essential to the capture of interdependencies which cannot be addressed by single-domain approaches. The AHP procedure of weight calculation allows expert or stakeholder decision to be factored in a systematic method and can be readily applied to participatory decision-making scenarios. Iterative sensitivity exploration, scenario assessment and repetition also have computational effectiveness to facilitate adaptive flood risk management.

4.4.3. Limitations and future research directions

Although the AHP weighting reflects expert judgment, a supplementary robustness evaluation combining AHP with two objective weighting methods (Entropy and CRITIC; Supplementary Table S8) confirms that PROMETHEE II rankings remain highly stable across all four alternative schemes ($\rho = 0.946\text{--}0.980$), while TOPSIS shows comparatively greater sensitivity ($\rho = 0.788\text{--}0.918$), consistent with the weight-sensitivity findings in Section 3.3.2. Future research could extend this comparison to stakeholder-elicited consensus weights to further validate participatory weighting strategies. Beyond Entropy and CRITIC, contemporary subjective weighting methods, such as the Best-Worst Method (Rezaei, 2015) and DEMATEL (Gabus and Fontela, 1972), offer a further avenue for comparison, given their respective advantages in reducing comparison burdens and modeling inter-criteria dependencies not captured by AHP.

All indicators had uniform preference thresholds (q and p) to make comparisons and ensure tractability. Individual threshold calibration can also be useful to improve discrimination and would be worth exploring, although the sensitivity analysis shows stability.

The analysis is cross-sectional. Longitudinal applications are required to assess temporal stability, resilience tracks, and the continuity of method concordance in changing conditions of hazards and development.

Observed flood damage is a limitation to external validation because it is not feasible to access high-resolution loss data. A combination of

post-event damage datasets would enhance empirical validation where possible.

To test generalizability, we extended the rank-reversal and weight-sensitivity tests (Tables 3–4) to MABAC (Pamućar and Ćirović, 2015) and SPOTIS (Dezert et al., 2020), implemented in crisp form consistent with the classical AHP weighting already used (Section 2.5; Tables S9–S10). Both methods show very high stability (mean weight-sensitivity $\rho = 0.9997$; rank-reversal rates of 0.02–0.83%), comparable to PROMETHEE II. However, both reduce mathematically to a simple weighted-sum aggregation ($\rho \approx 1.0000$ against an independently computed weighted-sum baseline), so their stability reflects linear insensitivity to extreme-point recalibration rather than PROMETHEE II's threshold-based outranking logic. This confirms TOPSIS's instability is specific to its ideal-point recalibration mechanism rather than a general weakness of distance-based aggregation, supporting our retained focus on the TOPSIS–PROMETHEE II comparison, two genuinely distinct aggregation philosophies, as the primary analysis. More extensive comparative analyses (ELECTRE, VIKOR, COPRAS, and group-decision variants of MCDA) would further inform the methodology in participatory flood resilience planning.

All in all, the research presents methodologically sound research on the relative agreement and consistency of two popular MCDA methods as a way of providing viable advice on flood resilience evaluation in highly risky, spatially explicit and constrained by data riverine settings.

5. Conclusion

The paper introduces a two-level framework of the comparative MCDA methodology in the flood resilience assessment which integrates concordance analysis with robustness validation analysis over 1000 spatial locations in the transboundary Brahmaputra River catchment of Kurigram District, Bangladesh. Based on 56 indicators of six dimensions of resilience (social, economic, institutional, infrastructural, environmental, and hydraulic) through AHP-TOPSIS and AHP-PROMETHEE II, a high overall concordance (Spearman's $\rho = 0.8741$; Kendall's $\tau = 0.6912$) was observed with 620 locations (62%) getting Nevertheless, non-trivial rank differences (mean = 112.3 positions; maximum = 464) indicate that there are methodological differences that should be carefully interpreted when prioritising interventions. The dimension-wise analysis also revealed a high concordance among the six dimensions of resilience ($\rho = 0.84\text{--}0.91$), such that the preference-based structure of PROMETHEE II and the distance-based measures of TOPSIS can provide broadly similar dimensional measurements, despite very different aggregation logics.

Intensive testing led to the discovery of performance differences between the two approaches that are critical. PROMETHEE II had outstanding stability; rank-reversal rates were eighty times lower (0.66% vs. 3.23%); and robustness to uncertainty in weights was nearly perfect (mean $\rho = 0.9986 \pm 0.0003$) with consistent rankings in the presence of perturbations in weights of up to 30 per cent. TOPSIS, in contrast, was highly sensitive to weight changes (mean $\rho = 0.0491 \pm 0.0035$), and all 56 indicators were highly sensitive, which is a significant drawback when using the indicator weights as expert judgement and when they are uncertain. The analysis of spatial autocorrelation (Global Moran = -0.0099 ; $p = 0.5541$) revealed that the ranking of disagreement is due to intrinsic aggregation processes and not due to geographic clustering. Concordance analysis at the indicator levels also showed that 96.4% of the indicators had very high concordance ($\Delta\rho < 0.1$), which shows that cumulative aggregation effects, not indicator instability, produce location-specific discordances.

In general, the suggested framework can help move flood resilience assessment to a higher level of dynamic and transparent characterisation of methodological robustness than the replication of correlation measures. Rather than asserting the universal superiority of either method, the results support conditional guidance grounded in the stability evidence presented: TOPSIS remains computationally efficient and

straightforward to interpret, making it suitable for exploratory analyses where indicator weights are reasonably well-established; PROMETHEE II's substantially greater robustness to weight and parameterization uncertainty makes it the more defensible choice specifically in decision contexts where weight elicitation is uncertain or contested and ranking stability carries direct operational consequences, such as resource allocation in high-stakes flood resilience planning. This guidance is based on robustness performance under the perturbation scenarios examined here, and the absence of external validation against observed flood damage means it should be treated as a methodological recommendation rather than a definitive decision rule.

This study provides a transferable evidence-based protocol to select the method of MCDA based on rigorous quantitative validation to support sound spatial planning and flood resilience in climate-exposed riverine systems across the globe. The proposed framework needs to be expanded to include temporal dynamics, real-time environmental monitoring, and cross-basin validation in order to further enhance its generalizability and policy relevance.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Grammarly/ChatGPT in order to language check and clarify the language of the paper. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Data availability statement

Data will be available based on reasonable request to the corresponding author.

CRediT authorship contribution statement

Md. Abdullah-AI Mamun: Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Mst. Nazneen Aktar:** Writing – review & editing, Investigation, Data curation. **Anjum Tasnuva:** Writing – review & editing, Resources, Data curation. **Md Tarikul Islam:** Writing – review & editing, Validation, Formal analysis. **Man- oranjan Mishra:** Writing – review & editing, Resources, Data curation. **Abu Reza Md. Towfiqul Islam:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

All authors affirm that they are neither a member nor have an interest of any kind in any organization or body in the content or subjects covered in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecohyd.2026.100812](https://doi.org/10.1016/j.ecohyd.2026.100812).

References

- Akter, S., Rahman, M.M., Moniruzzaman, M., 2025. Flood susceptibility analysis to sustainable development using MCDA and support vector machine models by GIS in the selected area of the Teesta River floodplain, Bangladesh. *HydroResearch* 8, 127–138. <https://doi.org/10.1016/j.hydres.2024.10.004>.
- Ansari, M.S., Warner, J., Sukhwani, V., Shaw, R., 2022. Implications of flood risk reduction interventions on community resilience: an assessment of community perception in Bangladesh. *Climate* 10 (2), 20. <https://doi.org/10.3390/cli10020020>.
- Anselin, L., 1995. Local indicators of spatial association—LISA. *Geogr. Anal.* 27 (2), 93–115. <https://doi.org/10.1111/j.1538-4632.1995.tb00338.x>.
- Aroca-Jiménez, E., Bodoque, J.M., García, J.A., 2023. An Integrated Multidimensional Resilience Index for urban areas prone to flash floods: development and validation. *Sci. Total. Environ.* 894, 164935.
- Bonferroni, C.E., 1936. Teoria statistica delle classi e calcolo delle probabilità. *Publ. R. Ist Super Sci. Econ. Commer. Firenze* 8, 3–62.
- Brans, J.-P., Vincke, P., 1985. A preference ranking organisation method: (The PROMETHEE method for multiple criteria decision-making). *Manag. Sci.* 31 (6), 647–656. <https://doi.org/10.1287/mnsc.31.6.647>.
- Çelen, A., 2014. Comparative analysis of normalization procedures in TOPSIS method: with an application to Turkish deposit banking market. *Informatica* 25 (2), 185–208. <https://doi.org/10.15388/Informatica.2014.10>.
- Chen, J., Gao, C., Zhou, H., Wang, Q., She, L., Qing, D., Cao, C., 2024. Urban flood risk assessment based on a combination of subjective and objective multi-weight methods. *Appl. Sci.* 14 (9), 3694. <https://doi.org/10.3390/app14093694>.
- Cochran, W.G., 1977. *Sampling Techniques*, 3rd ed. John Wiley & Sons, New York, NY.
- Coquelet, B., Dejaegere, G., De Smet, Y., 2025. Evaluating the PROMETHEE II ranking quality. *Algorithms* 18 (10), 597. <https://doi.org/10.3390/a18100597>.
- Debnath, P., Biswas, A., Jeong, G., 2025. Review of the use of the analytical hierarchy process for flood risk assessment in Bangladesh. *Nat. Hazards* 121, 19443–19472. <https://doi.org/10.1007/s11069-025-07523-6>.
- Dezert, J., Tchamova, A., Han, D., Tacnet, J.M., 2020. The SPOTIS rank reversal free method for multi-criteria decision-making support. In: 2020 IEEE 23rd International Conference on Information Fusion (FUSION), pp. 1–8. <https://doi.org/10.23919/FUSION45008.2020.9190347>.
- Diakoulaki, D., Mavrotas, G., Papayannakis, L., 1995. Determining objective weights in multiple criteria problems: the CRITIC method. *Comput. Oper. Res.* 22 (7), 763–770. [https://doi.org/10.1016/0305-0548\(94\)00059-H](https://doi.org/10.1016/0305-0548(94)00059-H).
- Donegan, H.A., Dodd, F.J., 1991. A note on Saaty's random indexes. *Math. Comput. Model.* 15 (10), 135–137. [https://doi.org/10.1016/0895-7177\(91\)90098-R](https://doi.org/10.1016/0895-7177(91)90098-R).
- Dutta, P., Deka, S., 2024. A novel approach to flood risk assessment: synergizing geospatial-based MCDM-AHP, multicollinearity, and sensitivity analysis in the Lower Brahmaputra floodplain, Assam. *J. Clean. Prod.* 467, 142985. <https://doi.org/10.1016/j.jclepro.2024.142985>.
- Edwards, W., Barron, F.H., 1994. SMARTs and SMARTER: improved simple methods for multiattribute utility measurement. *Organ. Behav. Hum. Decis. Process.* 60, 306–325. <https://doi.org/10.1006/obhd.1994.1087>.
- Efron, B., Tibshirani, R.J., 1994. *An Introduction to the Bootstrap*. Chapman and Hall/CRC. <https://doi.org/10.1201/9780429246593>.
- Ekmekcioğlu, Ö., Koc, K., Özger, M., 2021. Stakeholder perceptions in flood risk assessment: a hybrid fuzzy AHP-TOPSIS approach for Istanbul. *Turk. Int. J. Disaster Risk Reduct.* 60, 102327. <https://doi.org/10.1016/j.ijdrr.2021.102327>.
- Esmaili, R., Karipour, S.A., 2024. Comparison of weighting methods of multicriteria decision analysis (MCDA) in evaluation of flood hazard index. *Nat. Hazards* 120 (9), 8619–8638. <https://doi.org/10.1007/s11069-024-06541-0>.
- European Parliament & Council of the European Union, 2007. Directive 2007/60/EC of the European Parliament and of the Council of 23 October 2007 on the assessment and management of flood risks (Text with EEA relevance). *Off. J. Eur. Union* L 288, 27–34. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32007L0060>.
- Fawcett, T., 2006. An introduction to ROC analysis. *Pattern. Recognit. Lett.* 27 (8), 861–874. <https://doi.org/10.1016/j.patrec.2005.10.010>.
- Figueira, J., Roy, B., 2002. Determining the weights of criteria in the ELECTRE type methods with a revised Simos' procedure. *Eur. J. Oper. Res.* 139 (2), 317–326. [https://doi.org/10.1016/S0377-2727\(01\)00370-8](https://doi.org/10.1016/S0377-2727(01)00370-8).
- Fletcher, T.D., Shuster, W., Hunt, W.F., Ashley, R., Butler, D., Arthur, S., Trowsdale, S., Barraud, S., Semadeni-Davies, A., Bertrand-Krajewski, J.-L., Mikkelsen, P.S., Rivard, G., Uhl, M., Dagenais, D., Viklander, M., 2015. SUDS, LID, BMPs, WSUD and more: the evolution and application of terminology surrounding urban drainage. *Urban. Water J.* 12 (7), 525–542. <https://doi.org/10.1080/1573062X.2014.916314>.
- Foley, J.A., DeFries, R., Asner, G.P., Barford, C., Bonan, G., Carpenter, S.R., Chapin, F.S., Coe, M.T., Daily, G.C., Gibbs, H.K., Helkowski, J.H., Holloway, T., Howard, E.A., Kucharik, C.J., Monfreda, C., Patz, J.A., Prentice, I.C., Ramankutty, N., Snyder, P.K., 2005. Global consequences of land use. *Science* 309 (5734), 570–574. <https://doi.org/10.1126/science.1111772>.
- Folke, C., 2006. Resilience: the emergence of a perspective for social-ecological systems analyses. *Glob. Environ. Change* 16 (3), 253–267. <https://doi.org/10.1016/j.gloenvcha.2006.04.002>.
- Gabus, A., Fontela, E., 1972. *World problems: An invitation to Further Thought within the Framework of DEMATEL*. Battelle Geneva Research Centre.
- Haque, A., Shampa, Akter, M., Hussain, M.M., Rahman, M.R., Salehin, M., Rahman, M., 2024. An integrated risk-based early warning system to increase community resilience against disaster. *Prog. Disaster. Sci.* 21, 100310. <https://doi.org/10.1016/j.pdisas.2023.100310>.
- Haque, M.M., Islam, S., Sikder, M.B., Islam, M.S., 2022. Community flood resilience assessment in Jamuna floodplain: a case study in Jamalpur District, Bangladesh. *Int. J. Disaster. Risk. Reduct.* 72, 102861. <https://doi.org/10.1016/j.ijdrr.2022.102861>.
- Hossain, M.T., Haq, K.M.F., 2026. Flash flood risk delineation using multi-criteria decision-making (MCDM) approach in Sylhet Basin region, Bangladesh. *Total Environ. Adv.* 17, 200144. <https://doi.org/10.1016/j.teadv.2026.200144>.
- Hossain, M.N., Mumu, U.H., 2024. Flood susceptibility modelling of the Teesta River Basin through the AHP-MCDA process using GIS and remote sensing. *Nat. Hazards* 120, 12137–12161. <https://doi.org/10.1007/s11069-024-06677-z>.
- Hwang, C.-L., Yoon, K., 1981. *Multiple Attribute Decision making: Methods and applications—a State-Of-The-Art Survey*. Springer-Verlag, Berlin, Heidelberg. <https://doi.org/10.1007/978-3-642-48318-9>.
- IPCC, 2014. In: Field, C.B., et al. (Eds.), *Climate Change 2014: Impacts, Adaptation, and Vulnerability. Part A: Global and Sectoral Aspects. Contribution of Working Group II*

- to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press.
- Islam, A.R.M.T., Mamun, M.A., Hasan, M., Aktar, M.N., Uddin, M.N., Siddique, M.A.B., Senapathi, V., 2025. Optimizing coastal groundwater quality predictions: a novel data mining framework with cross-validation, bootstrapping, and entropy analysis. *J. Contam. Hydrol.* 269, 104480. <https://doi.org/10.1016/j.jconhyd.2024.104480>.
- Islam, A.R.M.T., Bappi, M.M.R., Alqadhi, S., Bindajam, A.A., Mallick, J., Talukdar, S., 2023. Improvement of flood susceptibility mapping by introducing hybrid ensemble learning algorithms and high-resolution satellite imageries. *Nat. Hazards* 119, 1–37. <https://doi.org/10.1007/s11069-023-06106-7>.
- Islam, A.R.M.T., Talukdar, S., Mahato, S., Kundu, S., Eibek, K.U., Pham, Q.B., Kuriqi, A., Linh, N.T.T., 2021. Flood susceptibility modelling using advanced ensemble machine learning models. *Geosci. Front* 12 (3), 101075. <https://doi.org/10.1016/j.gsf.2020.09.006>.
- Ji, J., Wang, D., 2023. Evaluation analysis and strategy selection in urban flood resilience based on EWM-TOPSIS method and graph model. *J. Clean. Prod.* 425, 138955. <https://doi.org/10.1016/j.jclepro.2023.138955>.
- Jenks, G.F., 1967. The data model concept in statistical mapping. *Int. Yearb. Cartogr.* 7, 186–190.
- Kersulienė, V., Zavadskas, E.K., Turskis, Z., 2010. Selection of rational dispute resolution method by applying new step-wise weight assessment ratio analysis (SWARA). *J. Bus. Econ. Manag.* 11 (2), 243–258. <https://doi.org/10.3846/jbem.2010.12>.
- Kizielewicz, B., Tomczyk, T., Gandor, M., Salabun, W., 2024. Subjective weight determination methods in multi-criteria decision-making: a systematic review. *Procedia Comput. Sci.* 246, 5396–5407. <https://doi.org/10.1016/j.procs.2024.09.673>.
- Liu, Y., Eckert, C.M., Earl, C., 2020. A review of fuzzy AHP methods for decision-making with subjective judgements. *Expert. Syst. Appl.* 161, 113738. <https://doi.org/10.1016/j.eswa.2020.113738>.
- Mamun, M.A.-A., Islam, A.R.M.T., Aktar, M.N., Uddin, M.N., Islam, M.S., Pal, S.C., Islam, A., Bari, A.B.M.M., Idris, A.M., Senapathi, V., 2024. Predicting groundwater phosphate levels in coastal multi-aquifers: a geostatistical and data-driven approach. *Sci. Total. Environ.* 953, 176024. <https://doi.org/10.1016/j.scitotenv.2024.176024>.
- Mann, H.B., Whitney, D.R., 1947. On a test of whether one of two random variables is stochastically larger than the other. *Ann. Math. Stat.* 18 (1), 50–60. <https://doi.org/10.1214/aoms/1177730491>.
- Millennium Ecosystem Assessment, 2005. *Ecosystems and Human well-being: Synthesis*. Island Press.
- Moghadas, M., Asadzadeh, A., Vafeidis, A., Fekete, A., Kötter, T., 2019. A multi-criteria approach for assessing urban flood resilience in Tehran, Iran. *Int. J. Disaster. Risk. Reduct.* 35, 101069. <https://doi.org/10.1016/j.ijdrr.2019.101069>.
- Mondal, M.S.H., Murayama, T., Nishikizawa, S., 2021. Examining the determinants of flood risk mitigation measures at the household level in Bangladesh. *Int. J. Disaster. Risk. Reduct.* 64, 102492. <https://doi.org/10.1016/j.ijdrr.2021.102492>.
- Pamućar, D., Čirović, G., 2015. The selection of transport and handling resources in logistics centers using multi-attribute border approximation area comparison (MABAC). *Expert. Syst. Appl.* 42 (6), 3016–3028. <https://doi.org/10.1016/j.eswa.2014.11.057>.
- Plataridis, K., Mallios, Z., 2024. Mapping flood susceptibility with PROMETHEE multi-criteria analysis method. *Environ. Sci. Pollut. Res.* 31 (28), 41267–41289. <https://doi.org/10.1007/s11356-024-33895-6>.
- Raff, J.L., Goodbred, S.L., Pickering, J.L., et al., 2023. Sediment delivery to sustain the Ganges-Brahmaputra delta under climate change and anthropogenic impacts. *Nat. Commun.* 14, 2429. <https://doi.org/10.1038/s41467-023-38057-9>.
- Rana, S.S., Habib, S.A., Sharifee, M.N.H., Sultana, N., Rahman, S.H., 2024. Flood risk mapping of the flood-prone Rangpur Division of Bangladesh using remote sensing and multi-criteria analysis. *Nat. Hazards Res.* 4 (1), 20–31. <https://doi.org/10.1016/j.nhres.2023.09.012>.
- Rezaei, J., 2015. Best-worst multi-criteria decision-making method. *Omega* 53, 49–57. <https://doi.org/10.1016/j.omega.2014.11.009>.
- Rizalilahi, M., Yulianur, A., Away, Y., Meilianda, E., 2026. GIS-based hybrid multi-criteria decision making in a fuzzy environment to delineate flood susceptibility regions in the Kr. Keureuto catchment area, North Aceh. *Results. Eng.* 29, 109064. <https://doi.org/10.1016/j.rineng.2026.109064>.
- Saaty, T.L., 1987. The analytic hierarchy process: what it is and how it is used. *Math. Model.* 9 (3–5), 161–176. [https://doi.org/10.1016/0270-0255\(87\)90473-8](https://doi.org/10.1016/0270-0255(87)90473-8).
- Saltelli, A., Tarantola, S., Campolongo, F., Ratto, M., 2004. *Sensitivity Analysis in practice: A guide to Assessing Scientific Models*. John Wiley & Sons, Chichester, UK. <https://doi.org/10.1002/0470870958>.
- Schär, S., Pohl, E., Geldermann, J., 2025. Analysing the compensatory properties of the outranking approach PROMETHEE. *J. Multi-Criteria Decis. Anal.* 32 (2), e70013. <https://doi.org/10.1002/mcda.70013>.
- Sheskin, D.J., 2011. *Handbook of Parametric and Nonparametric Statistical Procedures*, 5th ed. Chapman and Hall/CRC, New York, NY. <https://doi.org/10.1201/9780429186196>.
- Taoukidou, N., Karpouzou, D., Georgiou, P., 2025. Flood hazard assessment through AHP, fuzzy AHP, and frequency ratio methods: a comparative analysis. *Water* 17 (14), 2155. <https://doi.org/10.3390/w17142155>.
- Triantaphyllou, E., Sánchez, A., 1997. A sensitivity analysis approach for some deterministic multi-criteria decision-making methods. *Decis. Sci.* 28 (1), 151–194. <https://doi.org/10.1111/j.1540-5915.1997.tb01306.x>.
- UNDRR (United Nations Office for Disaster Risk Reduction), 2015. *Sendai Framework for Disaster Risk Reduction 2015–2030*. United Nations. <https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030>.
- Wang, X., Triantaphyllou, E., 2008. Ranking irregularities when evaluating alternatives by using some ELECTRE methods. *Omega* 36 (1), 45–63. <https://doi.org/10.1016/j.omega.2005.12.003>.
- Więckowski, J., Salabun, W., 2025. Comparative sensitivity analysis in composite material selection: evaluating OAT and COMSAM methods in multi-criteria decision-making. *Spectr. Mech. Eng. Oper. Res.* 2 (1), 24. <https://doi.org/10.31181/smeor21202524>.
- Zanakis, S.H., Solomon, A., Wishart, N., Dubliss, S., 1998. Multi-attribute decision-making: a simulation comparison of select methods. *Eur. J. Oper. Res.* 107 (3), 507–529. [https://doi.org/10.1016/S0377-2217\(97\)00147-1](https://doi.org/10.1016/S0377-2217(97)00147-1).